



Risk-averse two-stage distributionally robust optimisation for logistics planning in disaster relief management

Duo Wang^a, Kai Yang^b and Lixing Yang^a

^aState Key Laboratory of Rail Traffic Control and Safety, Beijing Jiaotong University, Beijing, People's Republic of China; ^bSchool of Traffic and Transportation, Beijing Jiaotong University, Beijing, People's Republic of China

ABSTRACT

Relief logistics is vital to disaster relief management. Herein, a risk-averse two-stage distributionally robust programming model is proposed to provide decision support for planning disaster relief logistics. It is distinct from the conventional disaster relief logistics planning problem in that (i) the facility location-inventory model and the multi-commodity network flow formulation are integrated; (ii) the probability distribution information of the supply, demand, and road link capacity is partially known, and (iii) the two-stage distributionally robust optimisation (DRO) method based on the worst-case mean-conditional value-at-risk criterion is developed. For tractability, we reformulate the proposed DRO model as equivalent mixed-integer linear programs for box and polyhedral ambiguity sets, which can be directly solved to optimality using the CPLEX software. To evaluate the validity of the proposed DRO model, we conduct numerical experiments based on a real-world case study addressing hurricane threats in the Gulf of Mexico region of the United States. Furthermore, we compare the performance of the proposed DRO model with that of the conventional two-stage stochastic programming model. Finally, we report the managerial implications and insights of using the risk-averse two-stage DRO approach for disaster relief management.

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1. Introduction

Earthquakes, landslides, floods, and wildfires are examples of disasters, which frequently occur worldwide and often result in casualties and significant social and economic losses. Owing to difficulty in predicting the time and position of disasters, a scientific and reasonable logistics planning method should be adopted for disaster relief management. As reported by Altay and Green (2006), disaster relief activities can be categorised into two sequential phases: preparation and response. Pre-disaster relief actions in the preparation stage typically include facility location and inventory pre-positioning for disaster relief items, whereas one of the post-disaster relief actions in the response stage is the distribution of relief items to demand sites. As pointed out by Ni, Shu, and Song (2018), considering these relief operations separately may result in sub-optimality. In this regard, the best solution for planning disaster relief logistics should account for these pre- and post-disaster actions simultaneously. To prevent sub-optimality, we optimise the facility location, pre-positioning inventory and distribution in the disaster relief logistics planning problem simultaneously.

The challenge in planning highly effective disaster relief logistics is the inherent uncertainty regarding the location, time, and scale of disasters (Barbarosolu and Arda 2004; Song, Chen, and Lei 2018). In particular, some key parameters in the disaster relief logistics planning problem, such as supply, demand, and road link capacity, are extremely uncertain. Under uncertain conditions, the disaster relief logistics planning problem satisfies a two-stage stochastic programming (SP) framework, in which the decisions of facility location and inventory correspond to the first stage, whereas relief supplies are allocated to satisfy the realised demand based on a multi-commodity network flow formulation in the second stage. A typical approach is to characterise the uncertainties using a finite number of scenarios, each of which occurs with a known probability. In practice, however, the amount of historical data is insufficient to obtain complete information regarding the probability distribution of uncertainties, particularly in a disaster environment. Therefore, based on the 'partial' and 'incomplete' information of uncertain parameters, the two-stage distributionally robust optimisation (DRO) method is a promising method for use in disaster relief logistics.

The conventional two-stage SP model minimises the deterministic first-stage cost and the expectation of the stochastic second-stage cost (Rawls and Turnquist 2010), which may yield average solutions; nevertheless, it may not be adequate for extreme realisations involving uncertain parameters. Additionally, in disaster relief management, decision-makers are typically risk-averse, and they do not favour risk-neutral approaches. Hence, risk associated with uncertainty must be considered when planning disaster relief logistics. Conditional value-at-risk (CVaR) is a widely used coherent risk indicator to avoid undesirable outcomes (Rockafellar and Uryasev 2000). As suggested by Noyan (2012), combining the expected cost and CVaR (i.e. mean-CVaR) can reduce the inefficiencies of the risk-neutral approach and the uncertainty risk simultaneously. However, the conventional mean-CVaR risk criterion lacks robustness (Zhu and Fukushima 2009). Therefore, we employ the worst-case mean-CVaR (WMCVaR) as a risk indicator to reflect decision-makers' risk-averse attitudes in the disaster relief logistics planning problem.

In summary, a novel risk-averse two-stage distributionally robust program is developed herein to formulate a disaster relief logistics planning problem under multi-dimensional uncertainty. The major contributions of this study are as follows.

- We propose a two-stage DRO model with the WMCVaR criterion by combining the facility location-inventory model and multi-commodity network flow formulation for the uncertain disaster relief logistics planning problem. To the best of our knowledge, this is the first study to adopt a hybrid modelling approach that requires only partial distribution information regarding uncertain supply, demand, and road link capacity for disaster relief management.
- We design two types of ambiguity sets (box and polyhedral) of possible probability distributions of uncertainties and convert the robust counterpart into equivalent computationally tractable formulations. Furthermore, we show the connection between the DRO model and the conventional two-stage SP model; subsequently, we prove that the proposed DRO model can be interpreted as a generalised version of the two-stage SP model.
- We conduct a realistic large-scale case study of hurricane threats in the southeastern United States to verify the effectiveness and applicability of the proposed DRO model. In addition, we compare the proposed DRO model with the conventional two-stage SP model and demonstrate the superiority of the former by performing an out-of-sample simulation analysis. By analysing the computational results, we obtain

managerial implications and insights regarding the use of the risk-averse two-stage DRO approach.

The rest of this paper is organised as follows. Section 2 reviews the relevant literature. Section 3 describes the studied problem. Section 4 presents the two-stage risk-averse DRO model for the disaster relief logistics planning problem and derives the equivalent deterministic formulations. Section 5 tests the proposed DRO model through a case study. Section 6 gives some concluding remarks and suggestions for future research.

2. Literature review

Our research is related to the optimisation methods for disaster relief logistics planning problem and the risk measures in the disaster relief logistics; thus, we review these issues in separate subsections. In the final subsection, we disclose the research gap and our main novelties.

2.1. Survey on optimisation methods for disaster relief logistics

The disaster relief logistics planning problem involves inherent uncertainties caused by the unpredictability of supply, demand, and transportation network availability (Yang et al. 2019; Zhang et al. 2021). This phenomenon has been widely observed. As reported by Tofghi, Torabi, and Mansouri (2016), the inherent uncertainties significantly affect the timely response of the relief organisations to disasters. In addition, ignorance of the uncertainties may result in inefficient logistics planning. Hence, it is critical to consider the uncertainties when making decisions on facility location, inventory pre-positioning, and relief delivery operations. The interested readers may refer to the survey of the uncertain disaster relief logistics planning problem in Galindo and Batta (2014), Anaya-Arenas, Renaud, and Ruiz (2014), Hoyos, Morales, and Akhavan-Tabatabaei (2015) and Grass and Fischer (2016).

The significance of uncertainty has prompted some researchers to use the SP method for the disaster relief logistics planning problem, where the uncertain parameters are described as random variables with exact probability distributions. For example, Rawls and Turnquist (2010) studied the location-inventory problem for pre-positioning relief items and formulated a two-stage SP, where the uncertain parameters of demands and capacities of network links were characterised by discrete joint probability distributions. Mete and Zabinsky (2010) established a SP model for disaster preparedness and response under uncertain demand. Garrido, Lamas, and Pino (2015) built a SP model to assist decision-makers

in the emergency logistics of a flood and attempted to optimise the levels of inventory in different zones and vehicles' availability. Rath, Gendreau, and Gutjahr (2016) employed a two-stage SP model to determine the locations for intermediate depots and the acquisition of vehicles and presented a transportation system for supplying disaster victims with relief items. Manopiniwes and Irohara (2017) presented a multi-objective SP model for the integrated decisions in the preparedness and response stages and applied their approach to a flood disaster in Chiang Mai province in northern Thailand. Krasko and Rebennack (2017) proposed a two-stage SP model for post-wildfire debris flow hazard management, which considered storm uncertainty and addressed the prevention of loss of life through emergency vehicle routing. Paul and Zhang (2019) proposed a two-stage SP model for the hurricane preparedness problem to determine the location of emergency response facilities, medical supply levels and transportation capacity. Subsequently, Wang (2020) introduced a two-stage SP model to evacuate the affected people and move them to safe areas during disaster response. Ghorbani and Ramezani (2020) developed a scenario-based two-stage SP to discuss joint decision-making regarding the selection of an integrated carrier for commodity and vehicle suppliers in an uncertain humanitarian supply chain.

Nonetheless, the above-mentioned studies assume complete information regarding the distribution of the uncertain variables, which is however not in line with the actual situations. The RO method limits the random parameters to given ambiguity sets and does not require any assumption regarding the probability distribution of uncertainties (Soyster 1973; Bertsimas and Sim 2004; Ben-Tal, Ghaoui, and Nemirovski 2009). Based on this, several researchers have introduced RO model to solve the disaster relief logistics planning problem (Bozorgi-Amiri, Jabalameli, and Mirzapour Al-e-Hashem 2013; Ni, Shu, and Song 2018; Paul and Wang 2019). For instance, Bozorgi-Amiri, Jabalameli, and Mirzapour Al-e-Hashem (2013) formulated a RO model for the disaster relief logistics planning problem by considering the expected value and the risk reflected by the variance of the total cost and the satisfaction of affected areas. Ni, Shu, and Song (2018) presented a RO model to address the uncertain parameters, including available pre-positioned inventories, post-disaster demand, and road capacities in the demand sites after a disaster occurs. Yahyaei and Bozorgi-Amiri (2019) developed a RO model to design a disaster relief logistics network for relief organisations to distribute commodities to demand sites in the aftermath of a disaster. Paul and Wang (2019) formulated two RO models that contained various earthquake magnitude-specific uncertainties including facility

damage, casualty by severity and travel time to obtain optimal logistics planning. Recently, Sarhadi, Naoum-Sawaya, and Verma (2020) presented a RO approach to model the problem of designing emergency response networks for marine oil-spills given the uncertainty in the location, size, and type of the spill.

Recent theoretical developments have revealed that the SP approach requires probability distributions of uncertain parameters, which are difficult to estimate accurately in reality, and the RO approach lacks any information about random parameters, which may result in over-conservative solutions. As an emerging approach, DRO can evaluate worst-case scenarios and overcome the inherent shortcomings of the two above-mentioned methods (Goh and Sim 2010; Wiesemann, Kuhn, and Sim 2014; Esfahani and Kuhn 2018; Huang et al. 2020; Ma, Liu, and Liu 2020; Jia, Liu, and Bai 2020). This is particularly new and has been gradually used to address the disaster relief logistics planning problem. For instance, Liu, Li, and Zhang (2019) introduced a DRO model to determine the location, number of ambulances and obtain an optimal demand assignment in an emergency medical service system by minimising the expected total cost. Wang and Chen (2020) presented a two-stage DRO model to optimise blood inventory pre-positioning and relief activities together, and described the uncertain distributions of blood demand by designing a moment-based ambiguous set. Zhang et al. (2020a) proposed a bi-objective DRO model for last-mile relief network under mixed transport to balance the safety and transportation time of relief networks.

2.2. Survey on risk measures in the disaster relief logistics

Most related existing works focus on adopting a risk-neutral criterion (i.e. the expectation of the second-stage optimal value) in the two-stage SP models for the disaster relief logistics planning problem (Rawls and Turnquist 2010, 2011; Rath, Gendreau, and Gutjahr 2016). Nevertheless, the risk-neutral method relying on the expectation may not be suitable for the reality of disaster relief systems. Moreover, the solutions obtained by the expected value models may be not robust due to high data variability. A common strategy used to study the risk-averse disaster relief logistics planning problem is to incorporate risk measures into the objective function (Noyan 2012; Alem, Clark, and Moreno 2016; Elçi and Noyan 2018). For example, Alem, Clark, and Moreno (2016) developed a two-stage stochastic network flow model to determine how to swiftly provide humanitarian aid for victims in the aftermath of the disaster and presented the computationally tractable methods of

state-of-art risk measures, such as semi-deviation (SD). Bai, Gao, and Liu (2018) proposed a value-at-risk (VaR) model for the pre-positioning emergency supplies problem through chance-constrained programming under different confidence levels.

It has been found that the conditional-value-at-risk (CVaR) is monotonic, transition-equivariant, convex, and positively homogeneous. For instance, Noyan (2012) proposed a risk-averse two-stage SP model and employed the CVaR as a risk indicator using a scenario optimisation method. Hu, Han, and Meng (2016) introduced a two-stage SP model to minimise financial budgets and risk-induced penalties, where the CVaR was employed as a decision-making tool to evaluate diverse decisions of prevention based on the degree of risk aversion. Elçi and Noyan (2018) presented a chance-constrained two-stage mean-risk SP model for the pre-disaster relief network design problem, which also used the CVaR as a risk measure. Zhong et al. (2020) developed a bi-objective mixed-integer nonlinear programming model for the disaster relief facility location and vehicle routing problem, which included conditional value at risk with regret (CVaR-R) as a risk measure. The research on the two-stage DRO with risk aversion for the disaster relief logistics planning constitutes a relatively new area. Zhang et al. (2020b) presented a DRO model and used mean absolute semi-deviation (MASD) as a risk indicator for humanitarian relief network design problem and transformed the chance constraint to its tractable formulation under a box+ellipsoid+generalised budget perturbation set.

2.3. Research gap and novelties

We classify the relevant literature in Table 1 to certify the novelties of our studied problem and modelling approach. As shown, no previous research has integrated

the WMCVaR risk indicator into a distributionally robust disaster relief logistics planning problem. Existing SP and RO researches assume that exact distributions and the bounded compactness of random parameters are provided, respectively. Hence, we select an intermediate approach between SP, which is inferior in terms of the distribution error; and RO, which performs conservatively on the support set of random variables. We develop a risk-averse two-stage DRO modelling framework for the disaster relief logistics planning problem under uncertainties based on multiple dimensions, including supply, demand, and road link capacity, to more effectively better utilise the limited distributional information to hedge against uncertainty.

To the best of our knowledge, the study by Zhang et al. (2020b) is the only other study that pertains to the distributionally robust disaster relief logistics planning problem. The differences between this study and that of Zhang et al. (2020b) are twofold: First, regarding the problem characteristics, this study captures both the node- and link-wise uncertainties including the demand in each affected area, proportion of usable inventory in each facility, and capacity of each road link in the disaster relief logistics network, whereas the problem proposed by Zhang et al. (2020b) considered only uncertain demand. Second, in terms of programming methods, a novel two-stage DRO model based on the WMCVaR criterion is developed in this study, whereas Zhang et al. (2020b) introduced a single-stage DRO model using the MASD risk indicator. Compared with the MASD, the WMCVaR can characterise the adjustable risk tolerance of the decision-maker and is more flexible in the DRO framework. In addition, we demonstrate that the WMCVaR-based DRO model can be equivalently transformed into a deterministic solvable linear program (LP) and performs competitively against the MASD-based model, thereby

Table 1. Summary of the most relevant studies.

Reference	Uncertainty			Distribution information			Methodology	Risk measure
	Supply	Demand	Road link capacity	Exact	Partially known	Unknown		
Rawls and Turnquist (2010)	✓	✓	✓	✓			SP	–
Mete and Zabinsky (2010)		✓		✓			SP	–
Noyan (2012)	✓	✓	✓	✓			SP	Mean-CVaR
Zokaee, Bozorgi-Amiri, and Sadjadi (2016)		✓				✓	RO	–
Alem, Clark, and Moreno (2016)	✓	✓		✓			SP	SD, CVaR
Rath, Gendreau, and Gutjahr (2016)		✓		✓			SP	–
Ni, Shu, and Song (2018)	✓	✓	✓			✓	RO	–
Elçi and Noyan (2018)	✓	✓		✓			SP	Mean-CVaR
Yahyaee and Bozorgi-Amiri (2019)		✓				✓	RO	–
Paul and Wang (2019)	✓	✓				✓	RO	–
Liu, Li, and Zhang (2019)		✓			✓		DRO	–
Wang (2020)		✓	✓	✓			SP	–
Ghorbani and Ramezani (2020)		✓		✓			SP	–
Zhang et al. (2020b)		✓			✓		DRO	MASD
Wang and Chen (2020)		✓			✓		DRO	–
This paper	✓	✓	✓		✓		DRO	WMCVaR

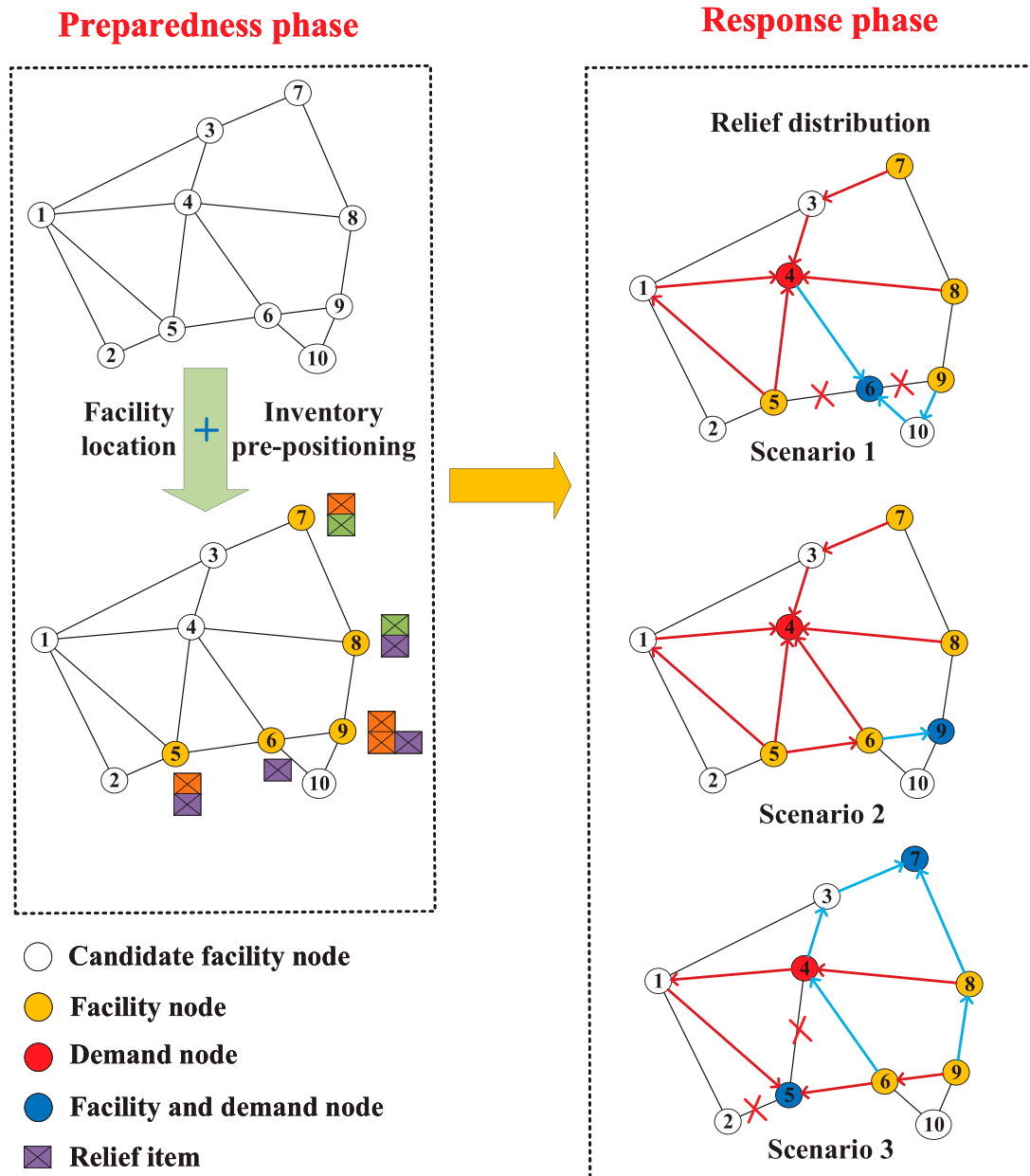


Figure 1. Illustration of a disaster relief logistics network.

increasing the computational challenge for nonlinear reformulations under the box and polyhedral ambiguity sets.

3. Problem description

Based on pre- and post-event actions, we consider a hierarchy of decisions for designing a disaster relief logistics network. In the first stage, the relief items are pre-positioned at the relief distribution centres before a disaster occurs, and then uncertainty is revealed, i.e. the random supply, demand, and road link capacity are observed. In the second stage, the available stocks

of different types of resources are transported across a transportation network to affected areas to provide immediate relief to disaster victims after the disaster occurs.

Consider a disaster relief logistics network with 10 nodes and 16 links, as shown in Figure 1, in which the red and yellow nodes represent demand and facility nodes, respectively, and the red cross symbol indicates that the transportation link is not available. Prior to a disaster, the locations of emergency facilities are determined in the first stage in the presence of uncertainty regarding post-disaster supply, demand, and network damage, as shown by nodes 5, 6, 7, 8, and 9 in Figure 1. In the aftermath of

a disaster, different distribution plans for relief supplies can be devised based on information received regarding the probability of uncertainties. For simplicity, three random scenarios with different demand nodes and road link capacities are considered to illustrate the problem. The transportation paths of certain relief supplies at different demand nodes are marked by red and blue lines under different scenarios. As shown in Figure 1 for the distribution plan under scenario 1, the demand in node 4 provided by different facilities is transported along paths $5 \rightarrow 4$, $5 \rightarrow 1 \rightarrow 4$, $7 \rightarrow 3 \rightarrow 4$, and $8 \rightarrow 4$, whereas the demand in node 6 is transported along paths $8 \rightarrow 4 \rightarrow 6$ and $9 \rightarrow 10 \rightarrow 6$. The demand in node 6 under scenario 1 cannot be supplied by the nearby facility 5; however, under scenario 2, the demand in nodes 4 and 9 can be supplied by facilities 5 and 6 along paths $5 \rightarrow 6 \rightarrow 4$ and $6 \rightarrow 9$, respectively. Paths $5 \rightarrow 6$ and $6 \rightarrow 9$ are available under scenario 2, but not available under scenario 1. Similarly, because path $4 \rightarrow 5$ is not available under scenario 3, the demand in node 5 under scenario 3 cannot be supplied by facility 8 along path $8 \rightarrow 4 \rightarrow 5$.

The aforementioned problem can be embedded into a two-stage modelling framework, where the first stage can be formulated using a facility location-inventory model, and the second stage distributes the supplies to satisfy the realised demand based on a multi-commodity network flow formulation. Without complete information regarding the probability distributions of uncertain parameters, we present a risk-averse two-stage distributionally robust programming to address the disaster relief logistics planning problem. The aim of this study is to determine the facility location, inventory, and delivery to minimise the deterministic total costs in the first stage, and to determine the WMCVaR of random total costs over member distributions arising from the ambiguity set in the second stage. The following are assumed to ease modelling.

Assumption 3.1: Emergency facilities that store relief items may be destroyed in a disaster, result in the partial loss of supplies stocked therein.

Assumption 3.2: In the aftermath of a disaster, certain sections of the transportation network used to transport relief items may become unavailable.

Assumption 3.3: Stocked relief items can only be allocated to open facilities, and those facilities must not stock relief items exceeding their available capacity.

Assumption 3.4: The demand node can only be satisfied by open emergency facilities.

4. Mathematical formulation

4.1. Notations and decision variables

To characterise the problem of interest, we first list all notations used in the mathematical formulation as follows:

Sets:

- N : the set of nodes
- I : the set of demand nodes, $I \subseteq N$
- K : the set of relief items
- L : the set of size categories of a facility
- A : the set of links in the network
- S : the set of all scenarios

Parameters:

- b^k : the unit space requirement for relief item k , $k \in K$
- F_l : the costs for opening a facility of size category $l \in L$ in location $i \in I$
- h^k : the unit salvage cost for unused relief item $k \in K$
- σ^k : the unit penalty cost for unsatisfied demand of relief item $k \in K$
- q^k : the unit acquisition cost for relief item $k \in K$
- u^k : the required link capacity for transporting a unit of relief item $k \in K$
- M_l : the maximal inventory capacity of a facility of size $l \in L$
- c_{ij}^k : the costs for shipping one unit of relief item $k \in K$ across link $(i, j) \in A$
- ϑ_i^{ks} : the realisation of demand for relief item $k \in K$ at location $i \in I$ in scenario $s \in S$
- ρ_i^{ks} : the realisation of the undamaged proportion of relief item $k \in K$ at node $i \in I$ in $s \in S$
- U_{ij}^s : the realisation of the maximal available capacity of link $(i, j) \in A$ in scenario $s \in S$
- $\xi^s = (\rho_i^{ks}, \vartheta_i^{ks}, \mathbf{U}^s)$: the realisation of the uncertain data in scenario s , $s \in S$

Decision variables:

- r_i^k : the amount of relief item $k \in K$ pre-positioned at location $i \in I$
- y_{il} : the binary variables, = 1 if a facility of size category $l \in L$ is opened at location $i \in I$; = 0, otherwise

- x_{ij}^{ks} : the amount of relief item $k \in K$ transported across link $(i, j) \in A$ in scenario $s \in S$
- z_i^{ks} : the amount of the unused relief item $k \in K$ at location $i \in I$ in scenario $s \in S$
- w_i^{ks} : the amount of the unsatisfied relief item $k \in K$ at location $i \in I$ in scenario $s \in S$

4.2. First stage: determination of emergency facility location and inventory level

Both the facility location and stocking decisions are made in the first stage, so opening facilities and pre-positioning relief supplies lead to cost on the first stage, i.e. prior to the disaster. Consequently, the objective of the first stage is to keep the total cost function at a minimum, which can be formulated as follows:

$$f(\mathbf{y}, \mathbf{r}, \omega) = \min \sum_{i \in I} \sum_{l \in L} F_l y_{il} + \sum_{k \in K} \sum_{i \in I} q^k r_i^k + \text{MCVaR}_\alpha(Q(\mathbf{y}, \mathbf{r}, \xi(\omega))), \quad (1)$$

where $Q(\mathbf{y}, \mathbf{r}, \xi(\omega))$ represents the random optimal value of the second-stage cost corresponding to the realisation of the random data $\xi(\omega)$ for the elementary event ω^s , denoted by $\xi^s = (\rho_i^{ks}, \vartheta_i^{ks}, \mathbf{U}^s)$. Here $\text{MCVaR}_\alpha(Q(\mathbf{y}, \mathbf{r}, \xi(\omega))) := (1 - \lambda)\mathbb{E}[Q(\mathbf{y}, \mathbf{r}, \xi(\omega))] + \lambda \text{CVaR}_\alpha(Q(\mathbf{y}, \mathbf{r}, \xi(\omega)))$, in which $\mathbb{E}$ represents the expected value operator, CVaR_α is the conditional value-at-risk with a given confidence level α and λ is the pre-specified weight of CVaR. In the following, we present the relevant definitions of the VaR and the CVaR.

Definition 4.1: Given a random variable Z , whose probability density function and cumulative distribution function are represented by $P_Z(\cdot)$ and $F_Z(\cdot)$, respectively. The VaR of Z , also called the α -quantile of Z , is defined as follows:

$$\text{VaR}_\alpha(Z) = \inf \{ \varphi \in \mathbb{R} : F_Z(\varphi) \geq \alpha \}. \quad (2)$$

Definition 4.2: Given a confidence level α , the CVaR of a random variable Z , denoted by $\text{CVaR}_\alpha(Z)$, is defined as

$$\text{CVaR}_\alpha(Z) = \min_{\varphi \in \mathbb{R}^+} \left\{ \varphi + \frac{1}{1 - \alpha} \mathbb{E}([Z - \varphi]_+) \right\}, \quad (3)$$

where $[a]_+ = \max(a, 0)$ represents the positive part of the real number a . As a measure of risk, CVaR captures a wide range of risk preferences and exhibits some better properties than VaR. For instance, Zhu and Fukushima (2009) showed that CVaR is a coherent risk measure, which is transition-equivariant, positively homogeneous, convex, and monotonic. Moreover, compared to VaR criterion, CVaR emphasises the average

level of excess losses and can reduce the unreliability aspect of risk variability.

Next, we group and explain the constraints involved in the first stage. Constraint (4) is used to ensure that relief items are allocated to open facilities and it is not allowed to stock more relief items in a facility than the available capacity. In addition, the number of open facilities at any node can not exceed one, which is limited by constraint (5). Both constraints (4) and (5) belong to the first stage and therefore do not depend on the scenario. It is noteworthy that the facility and pre-positioning decisions directly affect the second-stage decisions such as the number of transportation. Finally, the range of decision variables is restricted by constraints (6) and (7).

$$\sum_{k \in K} b^k r_i^k \leq \sum_{l \in L} M_l y_{il}, \quad \forall i \in I, \quad (4)$$

$$\sum_{l \in L} y_{il} \leq 1, \quad \forall i \in I, \quad (5)$$

$$y_{il} \in \{0, 1\}, \quad \forall i \in I, l \in L, \quad (6)$$

$$r_i^k \geq 0, \quad \forall i \in I, k \in K. \quad (7)$$

4.3. Second stage: distribution of relief supplies

According to the facility locations and inventory level pre-determined in the first stage, the second stage aims to generate an appropriate distribution planning under different scenarios. We allow infeasibilities in the stochastic constraints for all the scenarios, and hence introduce penalties on the amount of their violations. In this regard, we aim to achieve the following objective function, namely, the total costs associated with transportation, inventory holding and shortage are minimised. Therefore, by adopting a multi-commodity network flow formulation, the second-stage problem under the scenario s can be expressed as

$$Q(\mathbf{y}, \mathbf{r}, \xi^s) = \min \sum_{(i,j) \in A} \sum_{k \in K} x_{ij}^{ks} c_{ij}^k + \sum_{i \in I} \sum_{k \in K} (h^k z_i^{ks} + \sigma^k w_i^{ks}) \quad (8)$$

s.t.

$$\sum_{j \neq i \in N} x_{ji}^{ks} - \sum_{j \neq i \in N} x_{ij}^{ks} + \rho_i^{ks} r_i^k - z_i^{ks} + w_i^{ks} = \vartheta_i^{ks}, \quad \forall i \in N, k \in K, \quad (9)$$

$$\sum_{k \in K} u^k x_{ij}^{ks} \leq U_{ij}^s, \quad \forall (i, j) \in A, s \in S, \quad (10)$$

$$x_{ij}^{ks} \geq 0, \quad \forall (i, j) \in A, k \in K, s \in S, \quad (11)$$

$$z_i^{ks} \geq 0, w_i^{ks} \geq 0, \quad \forall i \in I, k \in K, s \in S. \quad (12)$$

The objective function (8) is to minimise the total costs associated with the transportation of the relief items to the affected areas, unsatisfied demand penalties and holding costs for unused supplies. The conservation of flow is represented by constraint (9). Constraint (10) is established to guarantee that the flow of each link cannot exceed the link capacity. In constraint (10), we have to consider an extreme case where the link may be unusable (i.e. $U_{ij}^s = 0$) in a specific scenario, and the relief items can not be delivered to the demand nodes. The non-negativity of decision variables are restricted by constraints (11) and (12).

4.4. Two-stage distributionally robust optimisation model

Assuming the probability distribution information of the random parameters are partially known, we formally presents the two-stage DRO model based on the WMCVaR criterion for the studied problem as follows

$$\begin{aligned} (\text{RA-TSDRO}) \quad & \min \sum_{i \in I} \sum_{l \in L} F_l y_{il} + \sum_{k \in K} \sum_{i \in I} q^k r_i^k \\ & + \text{WMCVaR}_\alpha(Q(\mathbf{y}, \mathbf{r}, \boldsymbol{\xi}(\omega))) \\ \text{s.t.} \quad & \text{constraints (4) - (12)}, \end{aligned} \quad (13)$$

in which $\text{WMCVaR}_\alpha(Q(\mathbf{y}, \mathbf{r}, \boldsymbol{\xi}(\omega))) = \sup_{\mathbf{p} \in \mathcal{P}} \text{MCVaR}_\alpha(Q(\mathbf{y}, \mathbf{r}, \boldsymbol{\xi}(\omega)))$ denotes the worst-case mean-CVaR of the second stage costs with respect to the distributional ambiguity set $\mathcal{P}$ containing the true distribution $\mathbf{p}$, $\mathbf{p} = (p_1, p_2, \dots, p_{|S|})^T$, $p_s > 0$ is the probability of scenario s and $\sum_{s \in S} p_s = 1$. The objective function (13) represents the following two goals in our proposed model, namely, (i) the cost resulting from opening facilities and acquiring relief supplies is minimised and (ii) the associated second stage cost has the minimum expectation and CVaR of the worst distribution from $\mathcal{P}$.

The proposed RA-TSDRO model is a semi-infinite programming model, which cannot be solved directly using optimisation softwares, e.g. CPLEX. To transform the aforementioned problem into a deterministic solvable optimisation model, we will design two types of ambiguity sets (box and polyhedral) of imprecise probability distributions of the uncertainties in the following.

4.4.1. Selection of ambiguity set of probability distributions

The ambiguity set of the distribution can flexibly characterise uncertainty by adopting partial information, which can be obtained from historical data or domain-specific knowledge. Two principles can be stipulated to select the ambiguity set: (1) $\mathcal{P}$ should contain the unknown true distribution with certainty (or at least, with high confidence); (2) $\mathcal{P}$ should facilitate a tractable (exact or safe approximate) formulation, such as LP. Among the various shapes of the ambiguity set that satisfy the two principles, we are particularly interested in the minimisation of WMCVaR under box and polyhedral uncertainties. This is because these two types of ambiguity sets are the easiest to specify, and the robust counterpart can be converted into a deterministic solvable optimisation problem (Zhu and Fukushima 2009; Ma, Liu, and Liu 2020). Hence, we constructed box and polyhedral ambiguity sets, to which $\mathbf{p}$ belongs, which are described by the partial probability distribution information:

$$\mathbf{p} \in \mathcal{P}_B = \{\mathbf{p} \mid \mathbf{p} = \mathbf{p}_0 + \boldsymbol{\zeta}, \mathbf{e}^T \boldsymbol{\zeta} = 0, \boldsymbol{\mu}^- \leq \boldsymbol{\zeta} \leq \boldsymbol{\mu}^+\}, \quad (14)$$

$$\begin{aligned} \mathbf{p} \in \mathcal{P}_\varphi = \{\mathbf{p} \mid \mathbf{p} = \mathbf{p}_0 + \mathbf{A}\boldsymbol{\zeta}, \mathbf{e}^T \mathbf{A}\boldsymbol{\zeta} = 0, \\ \mathbf{p}_0 + \mathbf{A}\boldsymbol{\zeta} \geq 0, \|\boldsymbol{\zeta}\|_1 \leq 1\}, \end{aligned} \quad (15)$$

where $\mathbf{p}_0$ denotes the nominal distribution of the discrete probabilities; $\mathbf{e} = (1, 1, \dots, 1)^T$; the perturbation vector $\boldsymbol{\zeta} = (\zeta_1, \zeta_1, \dots, \zeta_{|S|})^T$; and $\mathbf{A}$ is a known scaling matrix. Since the probability is non-negative, we can know that $\mathbf{p}_0 + \boldsymbol{\zeta} \geq 0$. In addition, the conditions $\mathbf{e}^T \boldsymbol{\zeta} = 0$ and $\mathbf{e}^T \mathbf{A}\boldsymbol{\zeta} = 0$ are to ensure that the vector $\mathbf{p}$ satisfies the basic condition of probability distribution. For instance, the scaling matrix can be set to $\mathbf{A} = \boldsymbol{\mu}^+ |S| \mathbf{I}$, where $\boldsymbol{\mu}^+$ represents the adjustable parameter and $\mathbf{I}$ is an identity matrix.

In addition, we focus on the rationality of the box and polyhedral ambiguity sets in practical disaster relief management applications. Decision-makers typically obtain a nominal or baseline estimation of the underlying probability distribution of the demand based on recorded disaster events. A simple method to design the distributional ambiguity is to generate a neighbourhood of the nominal probability distribution by allowing some perturbations around it. In other words, the decision-makers can determine the bounds for the box and the scaling matrix for polyhedral uncertainty based on behaved historical data or the prediction of experts. Theoretically, these two ambiguity sets can be formed with all probability distributions whose discrepancy to the nominal probability distribution is sufficiently small (Rahimian and Mehrotra 2019). In terms of application, the parameter $\boldsymbol{\mu}^+$ can quantify the discrepancy, and be interpreted as

the decision-makers' belief regarding the nominal probability distribution. In other words, as μ^+ decreases, the optimal value of the DRO model converges to an optimal value of the stochastic (nominal) problem (see Section 5 for details).

4.4.2. Reformulating the RA-TSDRO model

In this section, we turn the proposed RA-TSDRO model into its computationally tractable forms under the two ambiguity sets defined above. Notably, $\sup_{\mathbf{p} \in \mathcal{P}} \mathbb{E}_{\mathbf{p}}[Q(\mathbf{y}, \mathbf{r}, \xi(\omega))]$ and $\sup_{\mathbf{p} \in \mathcal{P}} \text{CVaR}_{\alpha}(Q(\mathbf{y}, \mathbf{r}, \xi(\omega)))$ in the RA-TSDRO model are related to the structure of the ambiguity set. According to the definition of the CVaR measure, the objective function of the RA-TSDRO model can be equivalently represented as follows:

$$\begin{aligned} & \min \sum_{i \in I} \sum_{l \in L} F_l y_{il} + \sum_{k \in K} \sum_{i \in I} q^k r_i^k \\ & + \sup_{\mathbf{p} \in \mathcal{P}} \min_{\varphi \in \mathbb{R}^+} \left\{ (1 - \lambda) \mathbb{E}_{\mathbf{p}}[Q(\mathbf{y}, \mathbf{r}, \xi(\omega))] \right. \\ & \left. + \lambda \left(\varphi + \frac{1}{1 - \alpha} \mathbb{E}_{\mathbf{p}}[(Q(\mathbf{y}, \mathbf{r}, \xi(\omega)) - \varphi)^+] \right) \right\}. \end{aligned} \quad (16)$$

Based on the desirable property of $Q(\mathbf{y}, \mathbf{r}, \xi(\omega))$, the order of the operators $\sup_{\mathbf{p} \in \mathcal{P}}$ and $\min_{\varphi}$ can be changed as follows:

$$\begin{aligned} & \sup_{\mathbf{p} \in \mathcal{P}} \min_{\varphi \in \mathbb{R}^+} \left\{ (1 - \lambda) \mathbb{E}_{\mathbf{p}}[Q(\mathbf{y}, \mathbf{r}, \xi(\omega))] \right. \\ & \left. + \lambda \left(\varphi + \frac{1}{1 - \alpha} \mathbb{E}_{\mathbf{p}}[(Q(\mathbf{y}, \mathbf{r}, \xi(\omega)) - \varphi)^+] \right) \right\} \\ & = \min_{\varphi \in \mathbb{R}^+} \sup_{\mathbf{p} \in \mathcal{P}} \left\{ (1 - \lambda) \mathbb{E}_{\mathbf{p}}[Q(\mathbf{y}, \mathbf{r}, \xi(\omega))] \right. \\ & \left. + \lambda \left(\varphi + \frac{1}{1 - \alpha} \mathbb{E}_{\mathbf{p}}[(Q(\mathbf{y}, \mathbf{r}, \xi(\omega)) - \varphi)^+] \right) \right\} \\ & = \min_{\varphi \in \mathbb{R}^+} \lambda \varphi + (1 - \lambda) \max_{\mathbf{p} \in \mathcal{P}} \mathbb{E}_{\mathbf{p}}[Q(\mathbf{y}, \mathbf{r}, \xi(\omega))] \\ & + \frac{\lambda}{1 - \alpha} \max_{\mathbf{p} \in \mathcal{P}} \mathbb{E}_{\mathbf{p}}[(Q(\mathbf{y}, \mathbf{r}, \xi(\omega)) - \varphi)^+]. \end{aligned} \quad (17)$$

Hence, we can reformulate the proposed RA-TSDRO model (13) as the following problem:

$$\begin{aligned} & \min \sum_{i \in I} \sum_{l \in L} F_l y_{il} + \sum_{k \in K} \sum_{i \in I} q^k r_i^k + \lambda \varphi \\ & + (1 - \lambda) \max_{\mathbf{p} \in \mathcal{P}} \mathbb{E}_{\mathbf{p}}[Q(\mathbf{y}, \mathbf{r}, \xi(\omega))] \\ & + \frac{\lambda}{1 - \alpha} \max_{\mathbf{p} \in \mathcal{P}} \mathbb{E}_{\mathbf{p}}[(Q(\mathbf{y}, \mathbf{r}, \xi(\omega)) - \varphi)^+] \\ & \text{s.t. constraints (4) - (12)}. \end{aligned} \quad (18)$$

By invoking the proof introduced by Ma, Liu, and Liu (2020), for the designed box and polyhedral uncertainty sets, we will convert the proposed models into their equivalent linear mixed-integer programming models in the following.

Theorem 4.3: *If probability $\mathbf{p}$ belongs to box ambiguity set (14), then RA-TSDRO model can be equivalently converted into the following deterministic solvable optimization model:*

$$\begin{aligned} & \min_{\mathbf{y}, \mathbf{r}, \varphi, \gamma, \tau, \eta, \gamma', \tau', \eta'} \sum_{i \in I} \sum_{l \in L} F_l y_{il} + \sum_{k \in K} \sum_{i \in I} q^k r_i^k \\ & + (1 - \lambda)(Q(\mathbf{y}, \mathbf{r}, \xi(\omega)))^T \mathbf{p}_0 + [\mu^-]^T \boldsymbol{\tau} + [\mu^+]^T \boldsymbol{\eta} \\ & + \lambda \left[\varphi + \frac{1}{1 - \alpha} (\mathbf{t}^T \mathbf{p}_0 + [\mu^-]^T \boldsymbol{\tau}' + [\mu^+]^T \boldsymbol{\eta}') \right] \\ & \text{s.t.} \begin{cases} t_s \geq Q(\mathbf{y}, \mathbf{r}, \xi^s) - \varphi, \forall s \in S, \\ \mathbf{e}\gamma + \boldsymbol{\tau} + \boldsymbol{\eta} = Q(\mathbf{y}, \mathbf{r}, \xi(\omega)), \\ \mathbf{e}\gamma' + \boldsymbol{\tau}' + \boldsymbol{\eta}' = \mathbf{t}, \\ t_s \geq 0, \forall s \in S, \\ \boldsymbol{\tau} \leq 0, \\ \boldsymbol{\eta} \geq 0, \\ \boldsymbol{\tau}' \leq 0, \\ \boldsymbol{\eta}' \geq 0, \\ \text{constraints (4) - (12)}, \end{cases} \end{aligned} \quad (19)$$

where $(\gamma, \gamma', \boldsymbol{\tau}, \boldsymbol{\tau}', \boldsymbol{\eta}, \boldsymbol{\eta}') \in R \times R \times R^{|S|} \times R^{|S|} \times R^{|S|} \times R^{|S|}$ are the dual variables/vectors.

Proof: Based on the definition of the expected value, the expected value of the second stage cost in (18) can be transformed into

$$\max_{\mathbf{p} \in \mathcal{P}} \mathbb{E}_{\mathbf{p}}[Q(\mathbf{y}, \mathbf{r}, \xi(\omega))] = \max_{\mathbf{p} \in \mathcal{P}} Q(\mathbf{y}, \mathbf{r}, \xi(\omega))^T \mathbf{p}. \quad (20)$$

In addition, by introducing an auxiliary variable $\mathbf{t} = (t_1, t_2, \dots, t_{|S|})^T$, $\max_{\mathbf{p} \in \mathcal{P}} \mathbb{E}_{\mathbf{p}}[(Q(\mathbf{y}, \mathbf{r}, \xi(\omega)) - \varphi)^+]$ can be equivalently expressed as

$$\begin{aligned} & \max_{\mathbf{p} \in \mathcal{P}} \mathbf{t}^T \mathbf{p} \\ & \text{s.t.} \begin{cases} t_s \geq Q(\mathbf{y}, \mathbf{r}, \xi^s) - \varphi, \forall s \in S, \\ t_s \geq 0, \forall s \in S. \end{cases} \end{aligned} \quad (21)$$

Assuming that probability distribution vector $\mathbf{p}$ belongs to the box ambiguity set (14), then problems (20) and (21) are both linear programming problems. Based on the definition of the expected value and box ambiguity set, $\sup_{\mathbf{p} \in \mathcal{P}} \mathbb{E}_{\mathbf{p}}[Q(\mathbf{y}, \mathbf{r}, \xi(\omega))]$ can be reformulated as:

$$\begin{aligned} & \sup_{\mathbf{p} \in \mathcal{P}_B} \mathbb{E}_{\mathbf{p}}[Q(\mathbf{y}, \mathbf{r}, \xi(\omega))] \\ & = \sup_{\mathbf{p} \in \mathcal{P}_B} Q(\mathbf{y}, \mathbf{r}, \xi(\omega))^T (\mathbf{p}_0 + \boldsymbol{\zeta}) \end{aligned}$$

$$\begin{aligned}
&= Q(\mathbf{y}, \mathbf{r}, \xi(\omega))^T \mathbf{p}_0 \\
&\quad + \max_{\xi} [Q(\mathbf{y}, \mathbf{r}, \xi(\omega))^T \xi \mid \mathbf{e}^T \xi = 0, \\
&\quad \mu^- \leq \xi \leq \mu^+].
\end{aligned} \tag{22}$$

According to the duality theory of linear programming, the dual form of problem (22) can be obtained as follows:

$$\begin{aligned}
&\min_{\gamma, \tau, \eta} Q(\mathbf{y}, \mathbf{r}, \xi(\omega))^T \mathbf{p}_0 + [\mu^-]^T \tau + [\mu^+]^T \eta \\
&\text{s.t.} \quad \begin{cases} \mathbf{e}\gamma + \tau + \eta = Q(\mathbf{y}, \mathbf{r}, \xi(\omega)), \\ \tau \leq 0, \\ \eta \geq 0. \end{cases}
\end{aligned} \tag{23}$$

Similarly, the equivalent form of the CVaR of the second stage cost function can be rewritten as

$$\begin{aligned}
&\max_{\mathbf{p} \in \mathcal{P}_B} \mathbf{t}^T (\mathbf{p}_0 + \xi) \\
&\text{s.t.} \quad \begin{cases} t_s \geq Q(\mathbf{y}, \mathbf{r}, \xi^s) - \varphi, \forall s \in S, \\ t_s \geq 0, \forall s \in S. \end{cases}
\end{aligned} \tag{24}$$

Note that the problem (24) is a linear programming problem about ξ , which can be formulated as

$$\begin{aligned}
&\max_{\xi} \xi^T \mathbf{t} \\
&\text{s.t.} \quad \begin{cases} \mathbf{e}^T \xi = 0, \\ \xi \geq \mu^-, \\ \xi \leq \mu^+. \end{cases}
\end{aligned} \tag{25}$$

The dual form of problem (24) is the following linear programming model:

$$\begin{aligned}
&\min_{\gamma', \tau', \eta'} \mathbf{t}^T \mathbf{p}_0 + [\mu^-]^T \tau' + [\mu^+]^T \eta' \\
&\text{s.t.} \quad \begin{cases} t_s \geq Q(\mathbf{y}, \mathbf{r}, \xi^s) - \varphi, \forall s \in S, \\ \mathbf{e}\gamma' + \tau' + \eta' = \mathbf{t}, \\ t_s \geq 0, \forall s \in S, \\ \tau' \leq 0, \\ \eta' \geq 0. \end{cases}
\end{aligned} \tag{26}$$

By combining models (18), (23) and (26), we can get the deterministic solvable formulation (19) of the RA-TSDRO model under the box ambiguity set. The proof is complete. ■

Theorem 4.4: If probability $\mathbf{p}$ belongs to polyhedral ambiguity set (15), then RA-TSDRO model can be equivalently

converted into the following deterministic solvable optimisation model:

$$\begin{aligned}
&\min_{\mathbf{y}, \mathbf{r}, \mathbf{q}, \nu, \pi, \mathbf{q}', \nu', \pi'} \sum_{i \in I} \sum_{l \in L} F_{il} y_{il} + \sum_{k \in K} \sum_{i \in I} q^k r_i^k \\
&\quad + (1 - \lambda) (Q(\mathbf{y}, \mathbf{r}, \xi(\omega))^T \mathbf{p}_0 + \mathbf{p}_0^T \mathbf{q} + \nu) \\
&\quad + \lambda \left[\varphi + \frac{1}{1 - \alpha} (\mathbf{p}_0^T \mathbf{t} + \mathbf{p}_0^T \mathbf{q}' + \nu') \right] \\
&\text{s.t.} \quad \begin{cases} t_s \geq Q(\mathbf{y}, \mathbf{r}, \xi^s) - \varphi, \forall s \in S, \\ \|\mathbf{A}^T Q(\mathbf{y}, \mathbf{r}, \xi(\omega)) + \mathbf{A}^T \mathbf{q} + \mathbf{A}^T \mathbf{e}\pi\|_{\infty} \leq \nu, \\ \|\mathbf{A}^T \mathbf{t} + \mathbf{A}^T \mathbf{q}' + \mathbf{A}^T \mathbf{e}\pi'\|_{\infty} \leq \nu', \\ t_s \geq 0, \forall s \in S, \\ \mathbf{q} \geq 0, \\ \nu \geq 0, \\ \mathbf{q}' \geq 0, \\ \nu' \geq 0, \\ \text{constraints (4) - (12),} \end{cases}
\end{aligned} \tag{27}$$

where $(\mathbf{q}, \mathbf{q}', \nu, \nu', \pi, \pi') \in R^{|S|} \times R^{|S|} \times R \times R \times R \times R$ are the dual variables/vectors.

Proof: Suppose probability distribution vector $\mathbf{p}$ belongs to a polyhedral ambiguity set (15), $\sup_{\mathbf{p} \in \mathcal{P}} \mathbb{E}_{\mathbf{p}}[Q(\mathbf{y}, \mathbf{r}, \xi(\omega))]$ can be represented as follows:

$$\begin{aligned}
&\sup_{\mathbf{p} \in \mathcal{P}_{\varphi}} \mathbb{E}_{\mathbf{p}}[Q(\mathbf{y}, \mathbf{r}, \xi(\omega))] \\
&= \sup_{\mathbf{p} \in \mathcal{P}_{\varphi}} Q(\mathbf{y}, \mathbf{r}, \xi(\omega))^T (\mathbf{p}_0 + \mathbf{A}\xi) \\
&= Q(\mathbf{y}, \mathbf{r}, \xi(\omega))^T \mathbf{p}_0 + \max_{\xi} [Q(\mathbf{y}, \mathbf{r}, \xi(\omega))^T \mathbf{A}\xi \mid \\
&\quad \mathbf{e}^T \mathbf{A}\xi = 0, \mathbf{p}_0 + \mathbf{A}\xi \geq 0, \|\xi\|_1 \leq 1] \\
&= Q(\mathbf{y}, \mathbf{r}, \xi(\omega))^T \mathbf{p}_0 + \Upsilon^*(Q),
\end{aligned} \tag{28}$$

where $\|\xi\|_1 = \sum_{s \in S} |\xi_s|$, and $\Upsilon^*(Q)$ is the optimal value of the following convex program

$$\begin{aligned}
&\max_{\xi} \{Q(\mathbf{y}, \mathbf{r}, \xi(\omega))^T \mathbf{A}\xi \mid \mathbf{e}^T \mathbf{A}\xi = 0, \\
&\quad \mathbf{p}_0 + \mathbf{A}\xi \geq 0, \|\xi\|_1 \leq 1\}.
\end{aligned} \tag{29}$$

The Lagrange function of problem (29) is

$$\begin{aligned}
&\mathcal{L}(\mathbf{q}, \nu, \pi, \xi) \\
&= Q(\mathbf{y}, \mathbf{r}, \xi(\omega))^T \mathbf{A}\xi + \mathbf{q}^T (\mathbf{p}_0 + \mathbf{A}\xi) \\
&\quad + \pi (\mathbf{e}^T \mathbf{A}\xi) + \nu (1 - \|\xi\|_1)
\end{aligned}$$

Subsequently, we obtain the following Lagrange dual function with variables $(\mathbf{q}, \nu, \pi) \in R^{|S|} \times R \times R$

$$g(\mathbf{q}, \nu, \pi) = \max_{\xi} \mathcal{L}(\mathbf{q}, \nu, \pi, \xi)$$

$$\begin{aligned}
&= (\mathbf{p}_0^T \boldsymbol{\varrho} + \nu) + \max_{\boldsymbol{\xi}} \{(\mathbf{A}^T Q(\mathbf{y}, \mathbf{r}, \boldsymbol{\xi}(\omega))) \\
&\quad + \mathbf{A}^T \boldsymbol{\varrho} + \mathbf{A}^T \mathbf{e}\pi\} \boldsymbol{\xi} - \nu \|\boldsymbol{\xi}\|_1\} \\
&= (\mathbf{p}_0^T \boldsymbol{\varrho} + \nu) + f^*(\mathbf{A}^T Q(\mathbf{y}, \mathbf{r}, \boldsymbol{\xi}(\omega))) \\
&\quad + \mathbf{A}^T \boldsymbol{\varrho} + \mathbf{A}^T \mathbf{e}\pi,
\end{aligned}$$

where

$$\begin{aligned}
&f^*(\mathbf{A}^T Q(\mathbf{y}, \mathbf{r}, \boldsymbol{\xi}(\omega))) + \mathbf{A}^T \boldsymbol{\varrho} + \mathbf{A}^T \mathbf{e}\pi \\
&= \begin{cases} 0, & \|\mathbf{A}^T Q(\mathbf{y}, \mathbf{r}, \boldsymbol{\xi}(\omega)) + \mathbf{A}^T \boldsymbol{\varrho} + \mathbf{A}^T \mathbf{e}\pi\|_\infty \leq \nu \\ \infty, & \text{otherwise} \end{cases}
\end{aligned}$$

is the conjugate function of $f(\boldsymbol{\xi}) = \nu \|\boldsymbol{\xi}\|_1$, and $\|\mathbf{A}^T Q(\mathbf{y}, \mathbf{r}, \boldsymbol{\xi}(\omega)) + \mathbf{A}^T \boldsymbol{\varrho} + \mathbf{A}^T \mathbf{e}\pi\|_\infty = \max_{s \in S} |(\mathbf{A}^T Q(\mathbf{y}, \mathbf{r}, \boldsymbol{\xi}(\omega)) + \mathbf{A}^T \boldsymbol{\varrho} + \mathbf{A}^T \mathbf{e}\pi)_s|$.

For any $\boldsymbol{\varrho} \geq 0$ and $\nu \geq 0$, the dual of problem (29) is given by

$$\begin{aligned}
&\min_{(\boldsymbol{\varrho}, \nu, \pi) \in R^{|S|} \times R \times R} \mathbf{p}_0^T \boldsymbol{\varrho} + \nu \\
&\text{s.t.} \quad \begin{cases} \|\mathbf{A}^T Q(\mathbf{y}, \mathbf{r}, \boldsymbol{\xi}(\omega)) + \mathbf{A}^T \boldsymbol{\varrho} + \mathbf{A}^T \mathbf{e}\pi\|_\infty \leq \nu, \\ \boldsymbol{\varrho} \geq 0, \\ \nu \geq 0. \end{cases}
\end{aligned} \tag{30}$$

Hence, the equivalent formulations of problem (28) regarding variables $\boldsymbol{\varrho}, \nu, \pi \in R^{|S|} \times R \times R$ can be written as

$$\begin{aligned}
&\min_{(\boldsymbol{\varrho}, \nu, \pi) \in R^{|S|} \times R \times R} Q(\mathbf{y}, \mathbf{r}, \boldsymbol{\xi}(\omega))^T \mathbf{p}_0 + \mathbf{p}_0^T \boldsymbol{\varrho} + \nu \\
&\text{s.t.} \quad \begin{cases} \|\mathbf{A}^T Q(\mathbf{y}, \mathbf{r}, \boldsymbol{\xi}(\omega)) + \mathbf{A}^T \boldsymbol{\varrho} + \mathbf{A}^T \mathbf{e}\pi\|_\infty \leq \nu, \\ \boldsymbol{\varrho} \geq 0, \\ \nu \geq 0. \end{cases}
\end{aligned} \tag{31}$$

Similarly, the equivalent form of the CVaR of the second stage cost function can be represented by

$$\begin{aligned}
&\min_{(\boldsymbol{\varrho}', \nu', \pi') \in R^{|S|} \times R \times R} \varphi + \frac{1}{1 - \alpha} (\mathbf{p}_0^T \mathbf{t} + \mathbf{p}_0^T \boldsymbol{\varrho}' + \nu') \\
&\text{s.t.} \quad \begin{cases} \|\mathbf{A}^T \mathbf{t} + \mathbf{A}^T \boldsymbol{\varrho}' + \mathbf{A}^T \mathbf{e}\pi'\|_\infty \leq \nu', \\ t_s \geq Q(\mathbf{y}, \mathbf{r}, \boldsymbol{\xi}^s) - \varphi, \forall s \in S, \\ t_s \geq 0, \forall s \in S, \\ \boldsymbol{\varrho}' \geq 0, \\ \nu' \geq 0. \end{cases}
\end{aligned} \tag{32}$$

By combining models (18), (31) and (32), we can obtain the deterministic solvable formulation (27) of the RA-TSDRO model under the polyhedral ambiguity set. The proof is complete. ■

Remark 4.1: As suggested by Venkatachalam and Narayanan (2019), the MASD risk indicator can be introduced into the objective function to achieve high data

variability. The MASD objective simultaneously considers the variability in the system performance and expected performance. Similar to the WMCVaR, we consider the MASD criterion for the distributionally robust disaster relief logistics planning problem, which is presented as follows:

$$\begin{aligned}
&\min \sum_{i \in I} \sum_{l \in L} F_l y_{il} + \sum_{k \in K} \sum_{i \in I} q^k r_i^k \\
&\quad + \text{MASD}(Q(\mathbf{y}, \mathbf{r}, \boldsymbol{\xi}(\omega))) \\
&\text{s.t.} \quad \text{constraints (4)-(12)},
\end{aligned} \tag{33}$$

where $\text{MASD}(Q(\mathbf{y}, \mathbf{r}, \boldsymbol{\xi}(\omega))) = \sup_{\mathbf{p} \in \mathcal{P}} \mathbb{E}_{\mathbf{p}}[\{Q(\mathbf{y}, \mathbf{r}, \boldsymbol{\xi}(\omega)) - \mathbb{E}_{\mathbf{p}}[Q(\mathbf{y}, \mathbf{r}, \boldsymbol{\xi}(\omega))]\}^+]$. The most obvious difference is that the MASD objective does not contain the confidence level α . Currently, the MASD measure cannot adjust the risk tolerance based on the risk-averse attitudes of decision-makers in planning disaster relief logistics. Mathematically, the deterministic equivalent formulations of the MASD model involve many nonlinear constraints under the box and polyhedral uncertainty sets, which are difficult to solve using commercially available solvers, such as CPLEX (see Appendix 1 for details).

Remark 4.2: It is important to highlight the relationship between our proposed two-stage DRO model and the conventional two-stage SP model. If the adjustment parameter $\boldsymbol{\mu}^+ = 0$, i.e. the box and polyhedral ambiguity sets reduce to the exact probability distribution, then the DRO models (19) and (27) are reduced to the conventional risk-averse two-stage SP model. In particular, for $\lambda = 0$ and $\boldsymbol{\mu}^+ = 0$, we can obtain the conventional risk-neutral two-stage SP model. This provides a meaningful relationship between the tractable robust counterpart forms and the conventional two-stage SP model.

5. Case study

In this section, a real-world case study addressing hurricane threats in the Gulf of Mexico region of the United States is presented to demonstrate the performance of the proposed DRO model and provide managerial implications and insights.

5.1. Data source

This case study is based on the study by Rawls and Turnquist (2010), which involves a network containing 30 nodes and 58 links. Owing to the size of the research network, Grass, Fischer, and Rams (2018) expanded the original network to 90 nodes by supplementing more realistic data, as shown in Figure 2. The data were from



Figure 2. Realistic case study network.

the International Best Track Archive for Climate Stewardship (IBTrACS) of the National Centers for Environmental Information, which records the data of hurricanes from 1900 to the present. For the case study, 900 counties in the Gulf of Mexico region affected by hurricanes were categorised into 90 nodes. It is noteworthy that in this case study, every node denotes the centre of the related region, and counties are deemed to be affected if they are within 80.5 km of a hurricane path.

For comparison, three sizes of case study instances, i.e. small, medium, and large, were designed as the problem sizes, as shown in Table 2. As depicted in Table 3, three different sizes of facilities: small, medium, and large can be opened with the construction cost and storage capacity, which are taken from Rawls and Turnquist (2010). For the small- and medium-sized case studies, three emergency commodities were considered: drinking water, food, and medicine. More specifically, the units of water were assumed to be 1000 gallons. Food was considered in the form of ready-to-eat meals (MREs), measured in units of 1000 meals. Medical kits designed for emergencies were considered in single units. The relief supplies delivered to the affected areas after the hurricanes are listed in Table 4. The specific space requirement, purchase price, and transportation costs for these commodities, as listed in Table 4, were obtained from Rawls and Turnquist (2010). In the large-scale case study, five kinds of commodities were considered: water, MREs, tent, medical set, and first aid kits (FAK). The unit purchase cost, storage volume occupied, and consumption share per unit of transport capacity listed in Table 4 are based on the South Carolina Emergency Operations Plan and the

Table 2. Problem sizes of different scale case studies.

	Small	Medium	Large
Scenarios	51	80	118
Nodes	30	90	90
Facility categories	3	3	3
Types of relief items	3	3	5
Constraints(Box)	12,198	670,260	1,009,788
Decision variables(Box)	167,760	1,988,223	4,886,631
Constraints(Polyhedral)	12,202	670,264	1,009,792
Decision variables(Polyhedral)	167,863	1,988,549	4,887,109

Table 3. Categories, setup cost and storage capacity of facilities in the case study (Rawls and Turnquist 2010).

Size category l	F_l (\$)	M_l (ft ³)
Small	19,600	36,400
Medium	188,400	408,200
Large	300,000	780,000

items catalog of the International Federation of Red Cross and Red Crescent Societies. Because of differences in the units, the unit purchase costs of the relief items are also different. According to Rawls and Turnquist (2010), the holding cost of unused items is one-quarter the resource purchase cost, and the unit penalty cost for unsatisfied demand is 10 times the purchase cost.

In the small-scale case study, as in Rawls and Turnquist (2010), 15 hurricanes were referenced to construct 51 scenarios to characterise the uncertain demand and network damage. In the medium- and large-scale case studies, the 80 and 118 scenarios used in Grass, Fischer, and Rams (2018) represent various demands for relief items and the destruction level of storage facilities and roads, respectively; the data were based on 203 hurricanes

Table 4. Unit purchase price, storage volume occupied and transportation costs for relief items in the small-, medium- and large-scale case studies (Grass, Fischer, and Rams 2018).

Case study	Commodity	q_k (\$/unit)	b_k (ft ³ /unit)	u_k (kg/unit)	Transport cost (\$/unit-mile)
Small	Water	647.7	144.6	144.6	0.3
	Food	5420	83.33	83.33	0.04
	Medical kits	140	1.16	1.16	0.00058
Medium	Water	0.68	0.093	1.13	0.5
	MRE	3.36	0.1029	0.76	0.5
	Tent	310	9.9499	55	0.5
Large	Water	0.68	0.093	1.13	0.5
	MRE	3.36	0.1029	0.76	0.5
	Tent	310	9.9499	55	0.5
	Medical set	770	6.3566	45	0.5
	FAK	870	2.2434	10.5	0.5

of different categories from 1900 to 2014. In these scenarios, any facility (and all relief items stocked therein) located at the landfall points are considered to be damaged, and the damage degree is proportional to the Saffir-Simpson Hurricane Scale. For minor hurricanes, 50% of the facilities and available relief items (if any) at the landfall point are damaged. For major hurricanes, any facility (and all relief items pre-positioned therein) located at the landfall points are considered to be destroyed. The occurrence probability of each scenario is calculated by performing an approximate matching of the aggregate historical characteristics of hurricanes recorded by the IBTrACS database. In particular, the probability of a scenario comprising several hurricanes is the sum of the probabilities of hurricanes occurring in that scenario. In this study, we obtained the nominal value of the scenario probability. For distributional ambiguity, the values of the adjustable parameter μ^+ in the box uncertainties were set as 0.001, 0.002, 0.003, and 0.004, separately.

The equivalent models for the DRO and corresponding SP can be solved directly using the CPLEX 12.10.1 optimisation software. For the medium- and large-scale case studies, the computation time limit was set to 20 min. All the numerical experiments were performed on a Huawei laptop comprising an Intel Core CPU (2.5 GHz) and 8 GB of memory.

5.2. Computational results for a real-world small-scale case study

5.2.1. Computational results and analysis

To illustrate the validity of the proposed DRO models under the two ambiguity sets previously defined, we tested the small-scale case study instances using $\alpha = 0.5$, $\lambda = 0.5$, and $\mu^+ = 0.001$. The location and amount of pre-positioned material are listed in Table A1 (see Appendix 2). As shown in Table A1, the number of facilities under the polyhedral uncertainty set exceeded that

under the box uncertainty set. The total amounts of pre-positioned relief supplies were 22327.832 and 22229.576 under the DRO models (19) and (27), respectively. The highest total pre-positioned amount was recorded by the distributionally robust model with the box ambiguity set. Under both uncertainty sets, only one medium-sized facility was assigned to store three different types of relief supplies. In addition, one of the emergency facilities stored only medical kits, whereas the remaining facilities stored two types of relief items. To analyse the effect of the confidence level on the location and pre-positioning plans, we further tested these two DRO models for $\alpha = 0.7$. As shown in Table A1, the location and scale of the facilities under the two different ambiguity sets did not differ significantly from the result based on $\alpha = 0.5$; however, the inventory level of each commodity increased significantly. Larger α values resulted in a more conservative decision, with higher inventory levels and lower shortage costs.

To illustrate the sensitivity of the solution to the penalty cost of unsatisfied demand for relief items, we conducted numerical experiments on a small-scale case study. More specifically, we increased the relative importance of water (the unfilled demand penalty cost for water increased by 10 times, reaching 100 times its purchase cost) and assumed that the penalty cost for insufficient water is κ times the purchase cost. The penalty costs for insufficient food and medical kits were set at 10 times their purchase cost. The resulting solutions for the location and storage amount of supplies are summarised in Table A1 (see Appendix 2). As shown, both the number of facilities and inventory levels increase significantly. Only the size of the facility located at node 30 was the same as that of the previous numerical result, and the combination of stored commodity differed. The other selected facilities were of a larger scale, and most of them stocked more water. Under the box and polyhedral uncertainty sets, the optimal solution allocated 20260.4306 and 20260.3306 units of water, respectively. Therefore, the amount of water reserved was sufficient to satisfy the demand; however, the increase in the penalty cost for water slightly decreased the amount of pre-positioned food and medicine.

To demonstrate the effect of the three parameters, λ , α , and μ^+ , we tested the small-scale case study instance under different confidence levels, risk-aversion parameters, and adjustment parameters. More specifically, we set the risk-aversion parameter $\lambda = 0, 0.2, 0.4, 0.6$, and 0.8 , and the confidence level $\alpha = 0.4, 0.6$, and 0.8 . We assumed that the adjustable parameter μ^+ was $0.001, 0.002, 0.003$, and 0.004 . The computational results for the box and polyhedral uncertainty sets are presented in Figure 3. As shown, the objective function value increased with the

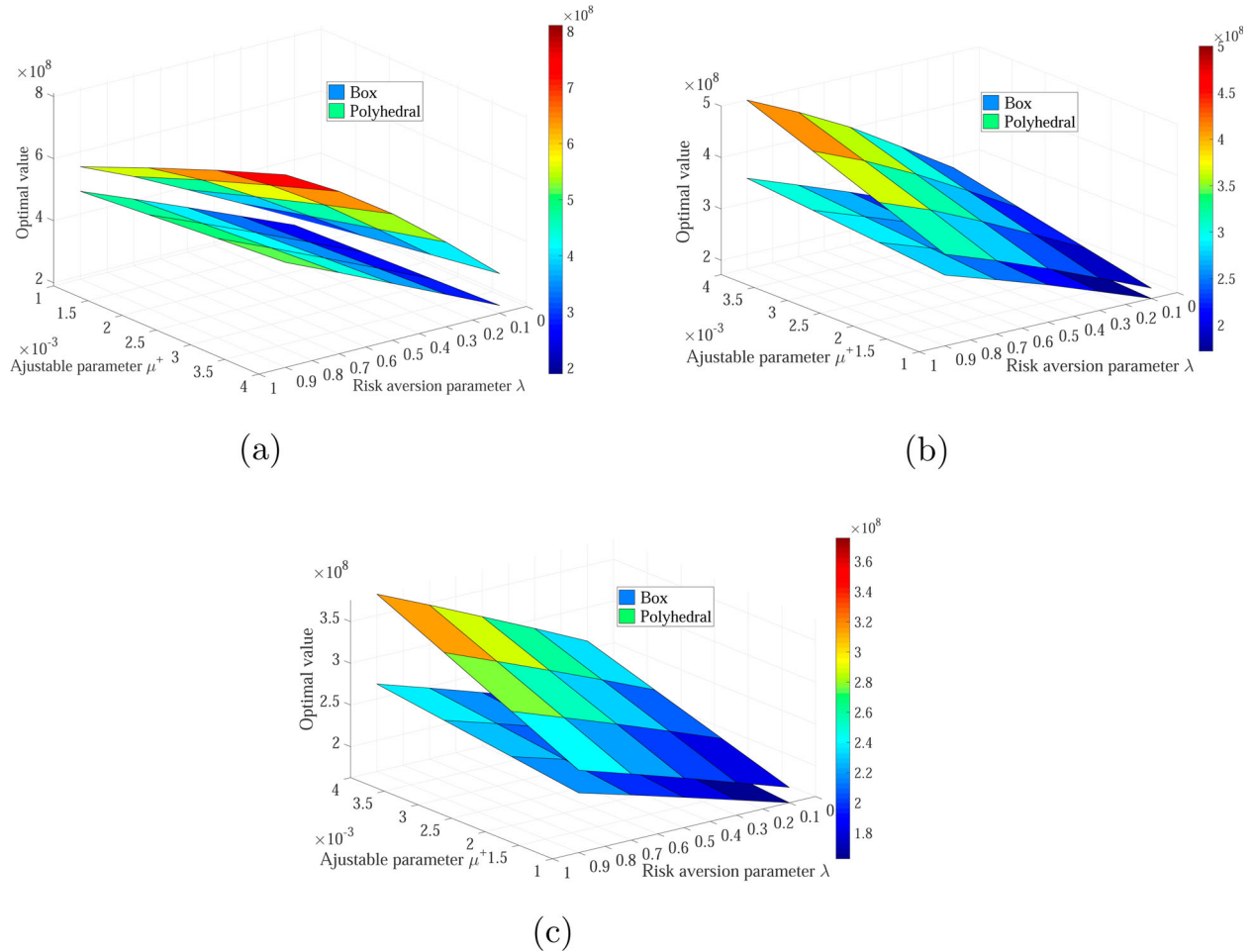


Figure 3. Comparisons of the DRO models under box and polyhedral ambiguity sets. (a) The optimal value for $\alpha = 0.8$. (b) The optimal value for $\alpha = 0.6$ and (c) The optimal value for $\alpha = 0.4$.

λ and α . Furthermore, the numerical results show that the optimal value of the DRO model increased with the adjustable parameter μ^+ . The change trend of the optimal value when $\alpha = 0.8$ was the same as those when $\alpha = 0.6$ and $\alpha = 0.4$, and they increased significantly with the adjustable parameters. In addition, it was observed that the objective function exhibited a linear relationship with parameters μ^+ and λ . Hence, we can conclude that these parameters significantly affect the optimal decisions; as such, in the actual disaster relief logistics planning, decision-makers must consider the confidence level α and perturbation parameter μ^+ . Additionally, as shown in Figure 3, the total cost under the two different ambiguity sets were different, which was larger under polyhedral ambiguity set. It can be confirmed that the box ambiguity set was smaller than the polyhedral ambiguity set. Namely, the box ambiguity set was completely enclosed by the polyhedral ambiguity set. Consequently, the polyhedral ambiguity set became larger and more conservative.

We further analysed the changes in the worst-case expected cost and CVaR for different values of risk parameters λ and α . Table 5 lists the comparison results of the relevant management performance indicators, i.e. the worst-case expected cost and CVaR with $\mu^+ = 0.001$. As shown in Table 5, the worst-case expected costs and CVaR of the DRO model under the polyhedral ambiguity set exceeded those under the box ambiguity set. When $\lambda = 0$, the objective function only contained some of the worst-case expected costs. The worst-case expected costs did not differ under different α values because they were not dependent on α . By contrast, when $\lambda = 1$, the objective function only contained the worst-case CVaR, which varied considerably with respect to parameter α .

Finally, we compared the optimal transportation routes for medicine under scenarios 6 and 24. We considered only the numerical results under the box uncertainty set as an example. The computational results are depicted in Figure 4, in which the red cross symbol indicates that the transportation link is not available, and

Table 5. Computational results of the worst-case expectation and CVaR.

Risk aversion parameter (λ)	Confidence level (α)	Model	Expected cost	CVaR $_{\alpha}$
0	0.6	Box	1.41×10^8	–
		Polyhedral	1.83×10^8	–
	0.8	Box	1.41×10^8	–
		Polyhedral	1.83×10^8	–
0.5	0.6	Box	7.14×10^7	1.60×10^8
		Polyhedral	9.43×10^7	2.13×10^8
	0.8	Box	7.14×10^7	2.51×10^8
		Polyhedral	9.43×10^7	3.59×10^8
1.0	0.6	Box	–	4.15×10^8
		Polyhedral	–	18.21×10^7
	0.8	Box	–	5.02×10^8
		Polyhedral	–	6.79×10^8

the green and yellow nodes represent the demand and facility nodes, respectively. Under different scenarios, the transportation paths for medicine at the demand node supplied by different facilities are indicated by coloured lines. As shown in Figure 4, the optimal transportation paths for medicine under two scenarios differed significantly. In particular, under scenario 6, the medicines supplied by facilities 5 and 22 were transported along paths $5 \rightarrow 10 \rightarrow 11 \rightarrow 12 \rightarrow 14 \rightarrow 15$, $22 \rightarrow 19 \rightarrow 18 \rightarrow 15$ and $22 \rightarrow 23 \rightarrow 26 \rightarrow 25 \rightarrow 24 \rightarrow 15$. Under scenario 24, the medicines supplied by facilities 5, and 22 were transported along paths $5 \rightarrow 10 \rightarrow 9 \rightarrow 8 \rightarrow 14$, $22 \rightarrow 19 \rightarrow 18 \rightarrow 15$ and $22 \rightarrow 23 \rightarrow 18 \rightarrow 16 \rightarrow 15$. By comparing these two scenarios, we discovered that scenario 6 contained more transportation paths than scenario 24; this is because most of the links connected to demand node 15 could not be used under scenario 24. Hence, we can conclude that the availability of transport networks significantly affects the design of disaster relief logistics networks.

5.2.2. Comparison results and analysis

We performed a comparative study between the proposed DRO model and the conventional two-stage SP model. The comparison results for different models based on $\alpha = 0.8$, $\lambda = 0.5$, and $\mu^+ = 0.001$ are listed in Table A2 (see Appendix 2). It can be observed that the location results differed significantly between the conventional SP and DRO models. For instance, the number of facilities in the SP model was two more than that in the DRO models under both ambiguity sets. Meanwhile, facilities that stored only one of the three commodity types outnumbered the other facilities. In addition, under the three different models, the inventory level of the facilities and the amount of different relief supplies stored were different. Therefore, it can be concluded that the optimal decisions were distinct in the disaster relief

logistics planning of the stochastic and DRO models. However, it can be easily seen that the optimal value of the DRO models were larger than that of the SP model because the probability distribution is exactly known in the stochastic model. Mathematically, the SP model can yield a theoretical optimal value; however, this is difficult to achieve in reality because of insufficient complete probability distribution information.

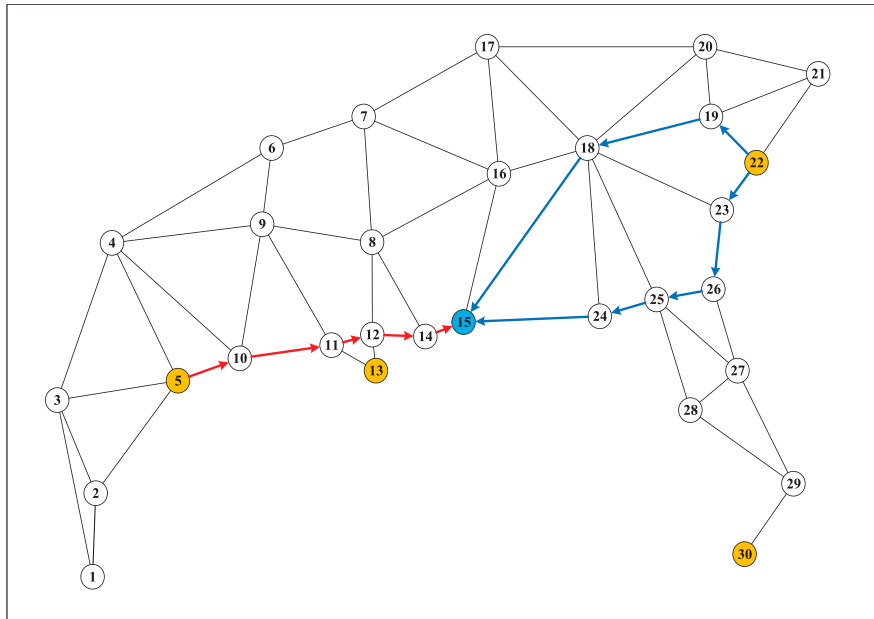
To further analyse the differences between the proposed DRO model and the two-stage SP model, we calculated the price of distributional robustness for the DRO model under the two different ambiguity sets. The price of distributional robustness (denoted as PDR) is defined as follows (Yin et al. 2019):

$$\text{PDR} = \frac{\text{DROV} - \text{TSOV}}{\text{TSOV}} \times 100\%,$$

in which TSOV and DROV represent the optimal values of the conventional SP and DRO models, respectively. The PDR represents the difference between the optimal values of the DRO and SP models relative to that of the SP model. As reported by Yin et al. (2019), although the DRO model needs to pay a small price, it avoids probability distribution ambiguity and uses probability distributional information more robustly than the conventional SP model with exact probabilistic distributions. According to the definition of PDR, we calculated the PDR for the DRO models using the two different ambiguity sets. Figure 5 shows the radar graph of the PDR under different adjustable parameters μ^+ with $\alpha = 0.8$ and $\lambda = 0.5$. As shown in the radar graph, the PDR became increasingly small as the adjustable parameter μ^+ decreased. This is because as the adjustment parameter decreased, the ambiguity set converged to the exact probability distribution, and the DRO model also degenerated to the conventional SP model. In addition, the maximum values of PDR under the box and polyhedral ambiguity sets did not exceed 10% and 46%, respectively. Hence, compared to box ambiguity set, the PDR for the DRO model under the polyhedral ambiguity set was greater; this implies that the larger the ambiguity set, the higher the cost the decision-makers need to bear. In this case involving incomplete probability distribution information, if the decision-makers adhere to the solution of the conventional two-stage SP model rather than the DRO model, mistakes may be generated. In summary, the DRO model can avoid the uncertainty of probability distributions by paying a small price.

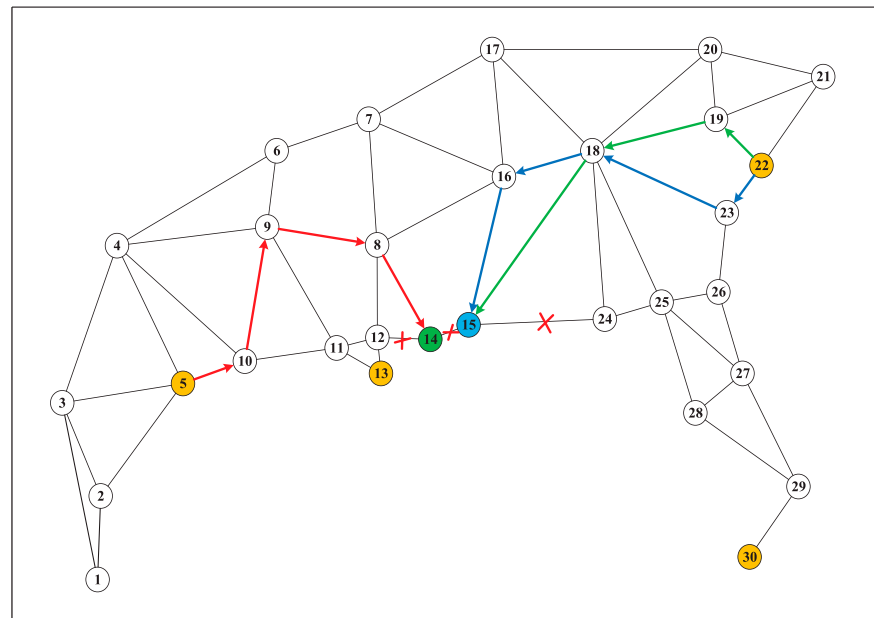
5.2.3. Out-of-sample simulation performance

We conducted a simulation study to compare the performances of our proposed DRO model and the SP model

Scenario 6:

● Facility and demand node

● Facility node

Scenario 24:

● Demand node

● Facility and demand node

● Facility node

Figure 4. Comparisons of optimal transportation routes under different scenarios.

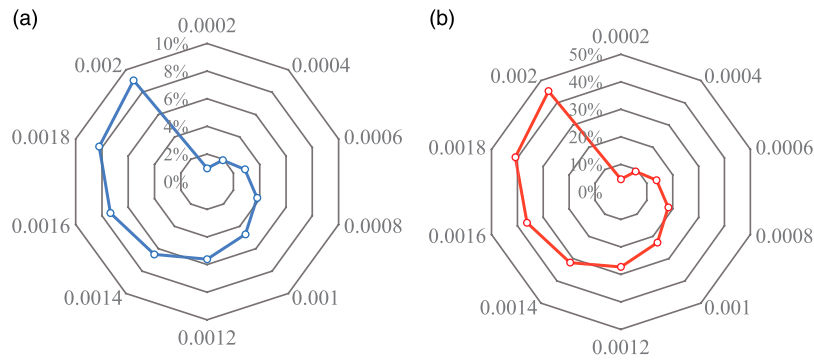


Figure 5. The radar graph of the PDR with different adjustable parameters. (a) Box and (b) Polyhedral.

Table 6. Out-of-sample performance of optimal solutions yielded by the DRO and SP models.

Adjustment parameter (μ^+)	Model	Metric			
		Mean	0.25-quantile	0.75-quantile	Worst-case
0.00001	SP	2,141,292,273	1,987,686,394	2,237,783,971	2,682,890,776
	Box	1,932,314,033	1,793,393,521	2,047,049,440	2,470,579,061
	Polyhedral	1,932,396,298	1,793,540,973	2,047,173,058	2,470,841,721
0.0001	SP	2,181,236,985	2,028,524,097	2,276,443,548	2,723,747,513
	Box	1,938,940,517	1,800,081,170	2,053,692,247	2,477,359,832
	Polyhedral	1,939,233,279	1,800,322,253	2,053,954,432	2,477,623,252

to obtain insights into the significance of robust decisions. We sampled $\{10, 50, 100\}$ dataset from a uniform distribution to test the out-of-sample performance of optimal solutions yielded by these abovementioned two models based on $\alpha = 0.8$, and $\lambda = 0.5$. More specifically, we first optimised these two models using the data provided and fixed the optimal first-stage decisions yielded by each model. Subsequently, we used the first-stage decisions to compute the out-of-sample cost under different adjustment parameters by solving the second-stage recourse problem. The out-of-sample results for the different models with a sample size of 100 are presented in Table 6. It was observed that the DRO model outperformed the SP model in terms of worst-case performance. This implies that the DRO model can effectively reduce the risk of the worst-case scenarios and is therefore highly desirable for risk-averse decision-makers who intend to hedge against extreme situations. In addition, as shown in Table 6, the differences between the out-of-sample performance of the DRO and SP models became increasingly small as the adjustable parameter μ^+ decreased. Figure 6 shows the mean (line) and (shaded) area between the 25% and 75% quantiles of the out-of-sample costs as a function of sample size. It is noteworthy that as the sample size increased, the out-of-sample costs of the DRO and SP models decreased, and the DRO models yielded significantly lower costs on average and at all quantiles compared with the SP model. In summary, the superior performance of the DRO reflects the significance of modelling the distributional ambiguity of uncertain parameters.

5.3. Computational results for real-world medium- and large-scale case studies

We conducted numerical experiments on real-world medium- and large-scale case studies to further investigate the effectiveness of our proposed DRO model. Because CPLEX cannot provide the optimal solution for large-scale case study rapidly, the obtained results are reported based on a 20 min limit. The computational results for different α and λ are listed in Table 7. As shown, the computation time increased significantly with the scale of the case study. In particular, for the large-scale case study, the gap obtained by using CPLEX was less than 1%, which demonstrates that the approximate solution can be accepted. Compared with the solutions obtained by Grass, Fischer, and Rams (2018), the location and inventory pre-positioning plan generated by the proposed DRO model based on the WMCVaR criterion performed more robustly.

5.4. Management implications and insights

In this section, we provide some managerial implications and insights for decision-makers based on the obtained computational results.

First, uncertainties abound in a realistic disaster relief logistics planning problem. This results in various risks that should not be disregarded; otherwise, an inefficient relief distribution will be yielded. Moreover, because the exact probability distribution information of the uncertain variables cannot be obtained in advance, decision-makers can only use partial information regarding the

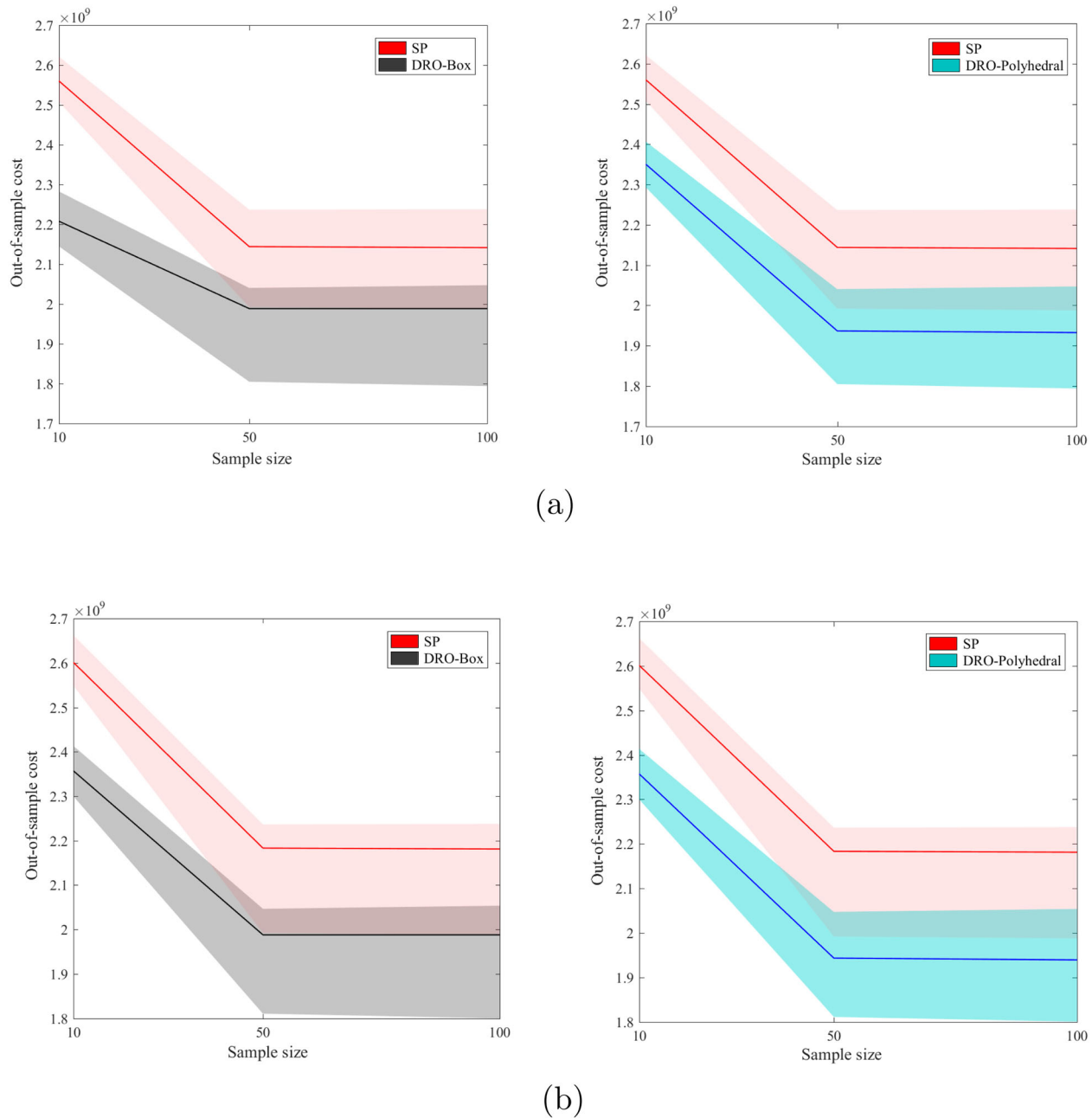


Figure 6. Out-of-sample performance under different adjustment parameters. (a) $\mu^+ = 0.00001$ and (b) $\mu^+ = 0.0001$.

probability distribution. Hence, we developed a risk-averse DRO model to provide decision support. The computational results indicate that overlooking the differences in the distribution information will result in both high decision-making risks and additional costs. It is note worthy that the proposed WMCVaR-based DRO model can avoid the risk caused by imprecise probability distribution information. Therefore, decision-makers can obtain reasonable logistics planning in disaster relief and hedge against uncertainties resulting from imprecise probability distribution information by employing the proposed DRO method.

Second, we applied the proposed DRO model to a real-world case study addressing hurricane threats in the Gulf of Mexico region of the United States. The results of out-of-sample simulation analysis show that the DRO model was superior to the SP model. The optimal decisions of the distributionally robust and conventional stochastic models differed significantly. This implies that the decision-makers should not disregard programming approach used for managing stochastic uncertainty in the optimal strategy. In fact, decision-makers should adopt more robust methods, such as distributionally robust programming. The computational results indicate

Table 7. Comparison results of the medium- and large-scale case studies.

Case study	Ambiguity set	α	λ	Optimal value	Gap	CPU time(s)
Medium-scale	Box	0.8	0.8	555,431,774.11	0.03%	333
		0.8	0.2	314,632,448.40	0.02%	343
		0.6	0.8	389,963,742.28	0.04%	393
	Polyhedral	0.8	0.8	683,546,534.01	0.00%	454
		0.8	0.2	390,562,799.89	0.01%	430
		0.6	0.8	470,845,603.39	0.00%	462
Large-scale	Box	0.8	0.8	1,490,588,680.29	0.02%	911
		0.8	0.2	870,494,751.01	0.00%	914
		0.6	0.8	1,045,130,736.36	0.02%	955
	Polyhedral	0.8	0.8	2,117,394,326.14	0.36%	1200
		0.8	0.2	1,223,488,813.46	0.58%	1200
		0.6	0.8	1,430,852,961.50	0.33%	1200

that the optimal objective value was dependent on the type of ambiguity set. In addition, λ and α significantly affected disaster relief logistics planning decisions. Increasing the λ and/or α implies a higher level of risk, which would result in more risk-averse plans. Therefore, compared with the MASD model, the proposed model allows decision-makers to select a parameter combination flexibly and evaluate the strategies based on their risk preferences.

6. Conclusion

In this study, a risk-averse two-stage DRO model was proposed to determine the locations of response facilities, emergency inventory pre-positioning and relief distribution of an entire disaster relief logistics network. This model simultaneously considered the uncertain demands, usable proportions of pre-positioned inventories, and road link capacity in the disaster relief logistics network due to the uncertain disaster environment. For the difficulty of model solvability caused by imprecise probability distributions, the tractable forms of the DRO models under the box and polyhedral ambiguity sets were deduced. Finally, the effectiveness and applicability of the proposed model were verified via a realistic large-scale case study addressing hurricane threats in the Gulf of Mexico region of the United States. The numerical results demonstrated that the proposed model performed well in terms of risk aversion and robustness.

However, the disaster relief process was considered to be static in this study; therefore, the dynamic relief items pre-positioning model should be developed in future studies. Moreover, the assumption that the relief resource has the same priority is another limitation. However, in practice, the priorities for different relief resource vary, and the rational deployment of disaster relief resources through priority can significantly improve their utilisation. Hence, in future studies, we will consider the priority of relief resources in the uncertain disaster relief logistics planning problem.

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Data availability statement

The data sets for the case study can be found at <https://zenodo.org/record/1012684>.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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Notes on contributors



Duo Wang received the B.S. and M.S. degrees in Mathematics from Henan University of Science and Technology, Luoyang, China, in 2016. She is currently a Ph.D. student in System Science from Beijing Jiaotong University. Her main research interests include emergency management, supply chain network design and distributionally robust optimisation. Her previous research has been published in *Evolution Equations and Control Theory*.



Kai Yang received the M.S. degree in Applied Mathematics from the Hebei University, Baoding, China, in 2012 and Ph.D. degree in Management Science and Engineering from the Tianjin University, Tianjin, China, in 2015. He is currently an Associate Professor at School of Traffic and Transportation, Beijing Jiaotong University, China. His interesting researches include stochastic

programming and robust optimisation, and their applications in transportation planning.



Lixing Yang received the B.S. and M.S. degrees from the Department of Mathematics, Hebei University, Baoding, China, in 1999 and 2002, respectively. In 2005, he received Ph.D. degree from the Department of Mathematical Sciences, Tsinghua University, Beijing, China. He is currently a Professor at State Key Laboratory of Rail Traffic Control and Safety, Beijing Jiaotong University, China. His current research interests include uncertain programming, intelligent systems, and applications in transportation planning and rail traffic control systems.

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Appendices

Appendix 1.

For the designed box and polyhedral uncertainty sets, we derive the deterministic solvable formulations for the MASD-based DRO model.

Theorem A.1: Under the box ambiguity set $\mathcal{P}_B$, the deterministic solvable formulation of the model (33) can be expressed as:

$$\begin{aligned} \min_{\mathbf{y}, \mathbf{r}, \phi, \varpi, \epsilon, \delta} \quad & \sum_{i \in I} \sum_{l \in L} F_l y_{il} + \sum_{k \in K} \sum_{i \in I} q^k r_i^k + \theta^T \mathbf{p}_0 + [\mu^-]^T \varpi \\ & + [\mu^+]^T \epsilon + \delta^T (Q(\mathbf{y}, \mathbf{r}, \xi(\omega)) - Q(\mathbf{y}, \mathbf{r}, \xi(\omega))^T \mathbf{p}_0 \mathbf{e} - \theta) \\ \text{s.t.} \quad & \begin{cases} \mathbf{e}\phi + \delta^T \mathbf{e}Q(\mathbf{y}, \mathbf{r}, \xi(\omega)) + \varpi + \epsilon = \theta, \\ \theta_s \geq 0, \forall s \in S, \\ \delta \leq 0, \\ \varpi \leq 0, \\ \epsilon \geq 0, \\ \text{constraints (4) – (12),} \end{cases} \end{aligned} \quad (\text{A1})$$

where $(\phi, \varpi, \epsilon, \delta) \in \mathbb{R} \times \mathbb{R}^{|S|} \times \mathbb{R}^{|S|} \times \mathbb{R}^{|S|}$ are the dual variables/vectors.

Proof: By introducing an auxiliary variable $\theta = (\theta_1, \theta_2, \dots, \theta_{|S|})^T$, $\max_{\mathbf{p} \in \mathcal{P}} \mathbb{E}_{\mathbf{p}}[Q(\mathbf{y}, \mathbf{r}, \xi(\omega)) - \mathbb{E}_{\mathbf{p}}[Q(\mathbf{y}, \mathbf{r}, \xi(\omega))]]^+$ can be equivalently expressed as

$$\begin{aligned} \max_{\mathbf{p} \in \mathcal{P}} \quad & \theta^T \mathbf{p} \\ \text{s.t.} \quad & \begin{cases} \theta_s \geq Q(\mathbf{y}, \mathbf{r}, \xi^s) - \mathbb{E}_{\mathbf{p}}[Q(\mathbf{y}, \mathbf{r}, \xi(\omega))], \forall s \in S, \\ \theta_s \geq 0, \forall s \in S. \end{cases} \end{aligned} \quad (\text{A2})$$

Assuming that probability distribution vector $\mathbf{p}$ belongs to the box ambiguity set (14), then problem (A2) can be

rewritten as

$$\begin{aligned} & \max_{\mathbf{p} \in \mathcal{P}_B} \boldsymbol{\theta}^T (\mathbf{p}_0 + \boldsymbol{\zeta}) \\ \text{s.t.} \quad & \begin{cases} \mathbf{e}^T \boldsymbol{\zeta} = 0, \\ \boldsymbol{\zeta} \geq \boldsymbol{\mu}^-, \\ \boldsymbol{\zeta} \leq \boldsymbol{\mu}^+, \\ \theta_s \geq Q(\mathbf{y}, \mathbf{r}, \xi^s) - Q(\mathbf{y}, \mathbf{r}, \xi(\omega))^T (\mathbf{p}_0 + \boldsymbol{\zeta}), \forall s \in S, \\ \theta_s \geq 0, \forall s \in S. \end{cases} \end{aligned} \quad (\text{A3})$$

Note that the problem (A3) is a linear programming problem about $\boldsymbol{\zeta}$, which can be formulated as

$$\begin{aligned} & \max_{\boldsymbol{\zeta}} \boldsymbol{\zeta}^T \boldsymbol{\theta} \\ \text{s.t.} \quad & \begin{cases} \mathbf{e}^T \boldsymbol{\zeta} = 0, \\ \boldsymbol{\zeta} \geq \boldsymbol{\mu}^-, \\ \boldsymbol{\zeta} \leq \boldsymbol{\mu}^+, \\ Q(\mathbf{y}, \mathbf{r}, \xi(\omega))^T \boldsymbol{\zeta} \geq Q(\mathbf{y}, \mathbf{r}, \xi^s) \\ \quad - Q(\mathbf{y}, \mathbf{r}, \xi(\omega))^T \mathbf{p}_0 - \theta_s, \forall s \in S. \end{cases} \end{aligned} \quad (\text{A4})$$

According to duality theory of linear programming, the dual form of problem (A3) is the following nonlinear programming model:

$$\begin{aligned} & \min_{\phi, \boldsymbol{\varpi}, \boldsymbol{\epsilon}, \delta} \boldsymbol{\theta}^T \mathbf{p}_0 + [\boldsymbol{\mu}^-]^T \boldsymbol{\varpi} + [\boldsymbol{\mu}^+]^T \boldsymbol{\epsilon} + \delta^T (Q(\mathbf{y}, \mathbf{r}, \xi(\omega)) \\ & \quad - Q(\mathbf{y}, \mathbf{r}, \xi(\omega))^T \mathbf{p}_0 \mathbf{e} - \boldsymbol{\theta}) \\ \text{s.t.} \quad & \begin{cases} \mathbf{e}\phi + \delta^T \mathbf{e}Q(\mathbf{y}, \mathbf{r}, \xi(\omega)) + \boldsymbol{\varpi} + \boldsymbol{\epsilon} = \boldsymbol{\theta}, \\ \theta_s \geq 0, \forall s \in S, \\ \delta \leq 0, \\ \boldsymbol{\varpi} \leq 0, \\ \boldsymbol{\epsilon} \geq 0. \end{cases} \end{aligned} \quad (\text{A5})$$

By combining models (33) and (A5), we can get the deterministic solvable formulation (A1). The proof is complete. ■

Theorem A.2: Under the polyhedral ambiguity set $\mathcal{P}_\phi$, the deterministic solvable formulation of the model (33) can be expressed as:

$$\begin{aligned} & \min_{\mathbf{y}, \mathbf{r}, \boldsymbol{\psi}, \boldsymbol{\zeta}, \boldsymbol{\epsilon}, \mathbf{v}} \sum_{i \in I} \sum_{l \in L} F_l y_{il} + \sum_{k \in K} \sum_{i \in I} q_i^k r_i^k + \boldsymbol{\theta}^T \mathbf{p}_0 + \mathbf{p}_0^T \boldsymbol{\psi} + \boldsymbol{\epsilon} \\ & \quad + \mathbf{v}^T [Q(\mathbf{y}, \mathbf{r}, \xi(\omega))^T \mathbf{p}_0 \mathbf{e} - Q(\mathbf{y}, \mathbf{r}, \xi(\omega)) + \boldsymbol{\theta}] \\ \text{s.t.} \quad & \begin{cases} \|\mathbf{A}^T \boldsymbol{\theta} + \mathbf{A}^T \boldsymbol{\psi} + \mathbf{A}^T \boldsymbol{\epsilon} \boldsymbol{\zeta} + \mathbf{A}^T \mathbf{e}Q(\mathbf{y}, \mathbf{r}, \xi(\omega))^T \mathbf{v}\|_\infty \leq \boldsymbol{\epsilon}, \\ \boldsymbol{\psi} \geq 0, \\ \mathbf{v} \geq 0, \\ \boldsymbol{\epsilon} \geq 0 \\ \theta_s \geq 0, \forall s \in S, \\ \text{constraints (4) - (12),} \end{cases} \end{aligned} \quad (\text{A6})$$

where $(\boldsymbol{\psi}, \boldsymbol{\zeta}, \boldsymbol{\epsilon}, \mathbf{v}) \in \mathbb{R}^{|S|} \times \mathbb{R} \times \mathbb{R} \times \mathbb{R}^{|S|}$ are the dual variables/vectors.

Proof: By introducing an auxiliary variable $\boldsymbol{\theta} = (\theta_1, \theta_2, \dots, \theta_{|S|})^T$, $\max_{\mathbf{p} \in \mathcal{P}} \mathbb{E}_{\mathbf{p}}[Q(\mathbf{y}, \mathbf{r}, \xi(\omega)) - \mathbb{E}_{\mathbf{p}}[Q(\mathbf{y}, \mathbf{r}, \xi(\omega))]]^+$ can be represented as follows:

$$\begin{aligned} & \max_{\mathbf{p} \in \mathcal{P}} \boldsymbol{\theta}^T \mathbf{p} \\ \text{s.t.} \quad & \begin{cases} \theta_s \geq Q(\mathbf{y}, \mathbf{r}, \xi^s) - \mathbb{E}_{\mathbf{p}}[Q(\mathbf{y}, \mathbf{r}, \xi(\omega))], \forall s \in S, \\ \theta_s \geq 0, \forall s \in S. \end{cases} \end{aligned} \quad (\text{A7})$$

Under the polyhedral ambiguity set, the model (A7) is given by

$$\begin{aligned} & \sup_{\mathbf{p} \in \mathcal{P}_\phi} \boldsymbol{\theta}^T (\mathbf{p}_0 + \mathbf{A}\boldsymbol{\zeta}) \\ & = \boldsymbol{\theta}^T \mathbf{p}_0 + \max_{\boldsymbol{\zeta}} [\boldsymbol{\theta}^T \mathbf{A}\boldsymbol{\zeta} \mid \mathbf{e}^T \mathbf{A}\boldsymbol{\zeta} = 0, \\ & \quad \mathbf{p}_0 + \mathbf{A}\boldsymbol{\zeta} \geq 0, \|\boldsymbol{\zeta}\|_1 \leq 1, \end{aligned} \quad (\text{A8})$$

$$\begin{aligned} & Q(\mathbf{y}, \mathbf{r}, \xi(\omega))^T \mathbf{A}\boldsymbol{\zeta} \\ & \geq Q(\mathbf{y}, \mathbf{r}, \xi^s) - Q(\mathbf{y}, \mathbf{r}, \xi(\omega))^T \mathbf{p}_0 - \theta_s, \forall s \in S \\ & = \boldsymbol{\theta}^T \mathbf{p}_0 + \Upsilon^*(\boldsymbol{\theta}), \end{aligned}$$

where $\|\boldsymbol{\zeta}\|_1 = \sum_{s \in S} |\zeta_s|$, and $\Upsilon^*(\boldsymbol{\theta})$ is the optimal value of the following convex program

$$\begin{aligned} & \max_{\boldsymbol{\zeta}} \boldsymbol{\theta}^T \mathbf{A}\boldsymbol{\zeta} \\ \text{s.t.} \quad & \begin{cases} \mathbf{e}^T \mathbf{A}\boldsymbol{\zeta} = 0, \\ \mathbf{p}_0 + \mathbf{A}\boldsymbol{\zeta} \geq 0, \\ \|\boldsymbol{\zeta}\|_1 \leq 1, \\ Q(\mathbf{y}, \mathbf{r}, \xi(\omega))^T \mathbf{A}\boldsymbol{\zeta} \\ \quad \geq Q(\mathbf{y}, \mathbf{r}, \xi^s) - Q(\mathbf{y}, \mathbf{r}, \xi(\omega))^T \mathbf{p}_0 - \theta_s, \forall s \in S. \end{cases} \end{aligned} \quad (\text{A9})$$

The Lagrange function of problem (A9) is

$$\begin{aligned} & \mathcal{L}(\boldsymbol{\psi}, \boldsymbol{\zeta}, \boldsymbol{\epsilon}, \mathbf{v}, \boldsymbol{\zeta}) \\ & = \boldsymbol{\theta}^T \mathbf{A}\boldsymbol{\zeta} + \boldsymbol{\zeta}^T (\mathbf{e}^T \mathbf{A}\boldsymbol{\zeta}) + \boldsymbol{\psi}^T (\mathbf{p}_0 + \mathbf{A}\boldsymbol{\zeta}) + \boldsymbol{\epsilon} (1 - \|\boldsymbol{\zeta}\|_1) \\ & \quad + \mathbf{v}^T [Q(\mathbf{y}, \mathbf{r}, \xi(\omega))^T \mathbf{A}\boldsymbol{\zeta} \mathbf{e} - Q(\mathbf{y}, \mathbf{r}, \xi(\omega)) \\ & \quad + Q(\mathbf{y}, \mathbf{r}, \xi(\omega))^T \mathbf{p}_0 \mathbf{e} + \boldsymbol{\theta}]. \end{aligned} \quad (\text{A10})$$

Then we obtain the following Lagrange dual function with variables $(\boldsymbol{\psi}, \boldsymbol{\zeta}, \boldsymbol{\epsilon}, \mathbf{v}) \in \mathbb{R}^{|S|} \times \mathbb{R} \times \mathbb{R} \times \mathbb{R}^{|S|}$

$$\begin{aligned} & g(\boldsymbol{\psi}, \boldsymbol{\zeta}, \boldsymbol{\epsilon}, \mathbf{v}) \\ & = \max_{\boldsymbol{\zeta}} \mathcal{L}(\boldsymbol{\psi}, \boldsymbol{\zeta}, \boldsymbol{\epsilon}, \mathbf{v}, \boldsymbol{\zeta}) \\ & = (\mathbf{p}_0^T \boldsymbol{\psi} + \boldsymbol{\epsilon} + \mathbf{v}^T [Q(\mathbf{y}, \mathbf{r}, \xi(\omega))^T \mathbf{p}_0 \mathbf{e} - Q(\mathbf{y}, \mathbf{r}, \xi(\omega)) + \boldsymbol{\theta}]) \\ & \quad + \max_{\boldsymbol{\zeta}} \{(\mathbf{A}^T \boldsymbol{\theta} + \mathbf{A}^T \boldsymbol{\psi} + \mathbf{A}^T \boldsymbol{\epsilon} \boldsymbol{\zeta} \\ & \quad + \mathbf{A}^T \mathbf{e}Q(\mathbf{y}, \mathbf{r}, \xi(\omega))^T \mathbf{v}) \boldsymbol{\zeta} - \boldsymbol{\epsilon} \|\boldsymbol{\zeta}\|_1\} \\ & = (\mathbf{p}_0^T \boldsymbol{\psi} + \boldsymbol{\epsilon} + \mathbf{v}^T [Q(\mathbf{y}, \mathbf{r}, \xi(\omega))^T \mathbf{p}_0 \mathbf{e} - Q(\mathbf{y}, \mathbf{r}, \xi(\omega)) + \boldsymbol{\theta}]) \\ & \quad + f^*(\mathbf{A}^T \boldsymbol{\theta} + \mathbf{A}^T \boldsymbol{\psi} + \mathbf{A}^T \boldsymbol{\epsilon} \boldsymbol{\zeta} + \mathbf{A}^T \mathbf{e}Q(\mathbf{y}, \mathbf{r}, \xi(\omega))^T \mathbf{v}), \end{aligned}$$

where $f^*(\mathbf{A}^T \boldsymbol{\theta} + \mathbf{A}^T \boldsymbol{\psi} + \mathbf{A}^T \boldsymbol{\epsilon} \boldsymbol{\zeta} + \mathbf{A}^T \mathbf{e}Q(\mathbf{y}, \mathbf{r}, \xi(\omega))^T \mathbf{v})$ is the conjugate function of $f(\boldsymbol{\zeta}) = \boldsymbol{\epsilon} \|\boldsymbol{\zeta}\|_1$.

Hence, the dual of problem (A9) is the following optimisation problem

$$\begin{aligned} & \min_{(\boldsymbol{\psi}, \boldsymbol{\zeta}, \boldsymbol{\epsilon}, \mathbf{v}) \in \mathbb{R}^{|S|} \times \mathbb{R} \times \mathbb{R} \times \mathbb{R}^{|S|}} \mathbf{p}_0^T \boldsymbol{\psi} + \boldsymbol{\epsilon} \\ & \quad + \mathbf{v}^T [Q(\mathbf{y}, \mathbf{r}, \xi(\omega))^T \mathbf{p}_0 \mathbf{e} - Q(\mathbf{y}, \mathbf{r}, \xi(\omega)) + \boldsymbol{\theta}] \\ \text{s.t.} \quad & \begin{cases} \|\mathbf{A}^T \boldsymbol{\theta} + \mathbf{A}^T \boldsymbol{\psi} + \mathbf{A}^T \boldsymbol{\epsilon} \boldsymbol{\zeta} \\ \quad + \mathbf{A}^T \mathbf{e}Q(\mathbf{y}, \mathbf{r}, \xi(\omega))^T \mathbf{v}\|_\infty \leq \boldsymbol{\epsilon}, \\ \boldsymbol{\psi} \geq 0, \\ \mathbf{v} \geq 0, \\ \boldsymbol{\epsilon} \geq 0. \end{cases} \end{aligned} \quad (\text{A11})$$

Hence, the equivalent programming of problem (A8) with regard to variables $(\boldsymbol{\psi}, \boldsymbol{\zeta}, \boldsymbol{\epsilon}, \mathbf{v}) \in \mathbb{R}^{|S|} \times \mathbb{R} \times \mathbb{R} \times \mathbb{R}^{|S|}$ can be

represented as follows:

$$\begin{aligned}
 & \min_{(\boldsymbol{\psi}, \boldsymbol{\zeta}, \varepsilon, \mathbf{v}) \in R^{|\mathcal{S}|} \times R \times R \times R^{|\mathcal{S}|}} \boldsymbol{\theta}^T \mathbf{p}_0 + \mathbf{p}_0^T \boldsymbol{\psi} + \varepsilon \\
 & + \mathbf{v}^T [Q(\mathbf{y}, \mathbf{r}, \boldsymbol{\xi}(\omega))^T \mathbf{p}_0 \mathbf{e} - Q(\mathbf{y}, \mathbf{r}, \boldsymbol{\xi}(\omega)) + \boldsymbol{\theta}] \\
 \text{s.t. } & \begin{cases} \|\mathbf{A}^T \boldsymbol{\theta} + \mathbf{A}^T \boldsymbol{\psi} + \mathbf{A}^T \mathbf{e}_{\boldsymbol{\zeta}} \\ + \mathbf{A}^T \mathbf{e} Q(\mathbf{y}, \mathbf{r}, \boldsymbol{\xi}(\omega))^T \mathbf{v}\|_{\infty} \leq \varepsilon, \\ \boldsymbol{\psi} \geq 0, \\ \mathbf{v} \geq 0, \\ \varepsilon \geq 0. \end{cases} \quad (\text{A12})
 \end{aligned}$$

By combining models (33) and (A12), we can get the deterministic solvable formulation (A6). The proof is complete. ■

Appendix 2.

For clarity of presentation, we list some long tables in Section 5 as follow.

Table A1. Location results and storage amount of supplies under different confidence levels and penalty costs.

α	κ	Ambiguity set	City	Facility size	Water(1000gals)	Food(1000gals)	Medical kits
0.5	10	Box	5	Medium	2065.7	1125	13,578
			13	Small	228.71	39.947	0
			15	Small	0	48.002	3448.3
			22	Small	0	57.783	880.14
			30	Small	0	12.252	844
			Total =		2294.41	1282.982	18,750.44
		Polyhedral	5	Medium	2180.5	925.9	13,578
			9	Small	0	0	578.66
			13	Small	228.71	39.947	0
			15	Small	0	54.997	2945.8
			22	Small	0	36.81	804
			30	Small	0	12.252	844
			Total =		2409.21	1069.906	18,750.46
0.7	10	Box	5	Medium	2065.7	1125	13,578
			13	Small	228.71	39.947	0
			15	Small	0	175.32	2155.9
			22	Small	118	61.589	1863.4
			30	Small	0	0	1724.1
			Total =		2412.41	1401.856	19,321.4
		Polyhedral	5	Medium	2230	839.91	13578
			9	Small	0	0.78675	522.14
			13	Small	228.71	39.947	0
			15	Small	0	191.02	2045
			22	Small	0	49.885	2813
			30	Small	0	12.252	844
			Total =		2458.71	1133.80075	19,802.14
0.7	100	Box	5	Large	2250	1067.5	13,578
			11	Medium	2823	0	0
			13	Medium	2802.1	36.267	0
			15	Large	5372.3	0	2726.9
			19	Small	165.84	0	0
			20	Small	251.73	0	0
			21	Medium	1200	48.002	0
			22	Large	5388.4	0	721.38
			30	Small	7.0606	0	1724.1
			Total =		20,260.4306	1151.769	18,750.38
		Polyhedral	5	Medium	2250	805.23	13,578
			11	Medium	2823	0	0
			13	Medium	2809.1	24.001	0
			15	Large	5365.2	12.266	2726.9
			19	Small	165.84	0	0
			20	Small	251.73	0	0
			21	Medium	1200	48.002	0
			22	Large	5388.4	0	721.38
			30	Small	7.0606	0	1724.1
			Total =		20,260.3306	889.499	18,750.38

Table A2. Comparison results of small-scale case study.

Model	Optimal value	City	Facility size	Water(1000gals)	Food(1000gals)	Medical kits
Stochastic	318,645,470.58	5	Medium	2065.7	1125	13,578
		6	Small	0	0	920.45
		13	Small	2228.71	39.947	0
		15	Small	0	30.421	2644.3
		21	Small	0	48.002	0
		22	Small	0	36.81	804
		30	Small	0	12.238	845
			Total =	2294.41	1292.418	18,791.75
Box	333,659,261.05	5	Medium	2065.7	1125	13,578
		9	Small	0	8.6984	1099.3
		13	Small	4.642	39.947	0
		15	Small	0	50.496	3269.1
		22	Small	0	36.81	804
		30	Small	0	9.3604	1051.7
			Total =	2070.342	1270.3118	19,802.1
Polyhedral	391,291,321.43	5	Medium	2190.7	1049.1	3448.3
		9	Small	0	8.0552	1145.5
		13	Small	228.71	39.947	0
		15	Small	0	63.083	2364.9
		22	Small	0	0.86499	3386.1
		30	Small	0	12.252	844
			Total =	2419.41	1173.30219	11,188.8

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